

Statistics  
367-1-4361  
Normal Models  
Opher Donchin

# 7.1 Continuous data

Types of data, normal model, multivariate likelihood, graphical representation

# Likelihoods of biomedical data

- All data in lectures so far has been binary
  - Tutorial has done counts as well
- Likelihoods have had only one parameter
- Changing the type of data requires changing the likelihood

| Binary            | Grouped       | Ordered Groups         | Counts          | Continuous         |
|-------------------|---------------|------------------------|-----------------|--------------------|
| Survived / died   | Blood type    | Pain score             | Number of falls | Reaction time (ms) |
| Patient / control | Cell type     | Sleep quality          | Spike count     | Voice pitch (Hz)   |
| Mutation present  | Diagnosis     | Disease stage          | # of seizures   | Cerebellar volume  |
| Drug responder    | Sleep stage   | Tumor grade            | # of mutations  | Blood pressure     |
| Seizure occurred  | Strategy type | Confidence rating      | # of infections | Movement error     |
| Binomial          | Multinomial   | Ordered logistic model | Poisson         | Normal             |

# A database of anatomical brain scans

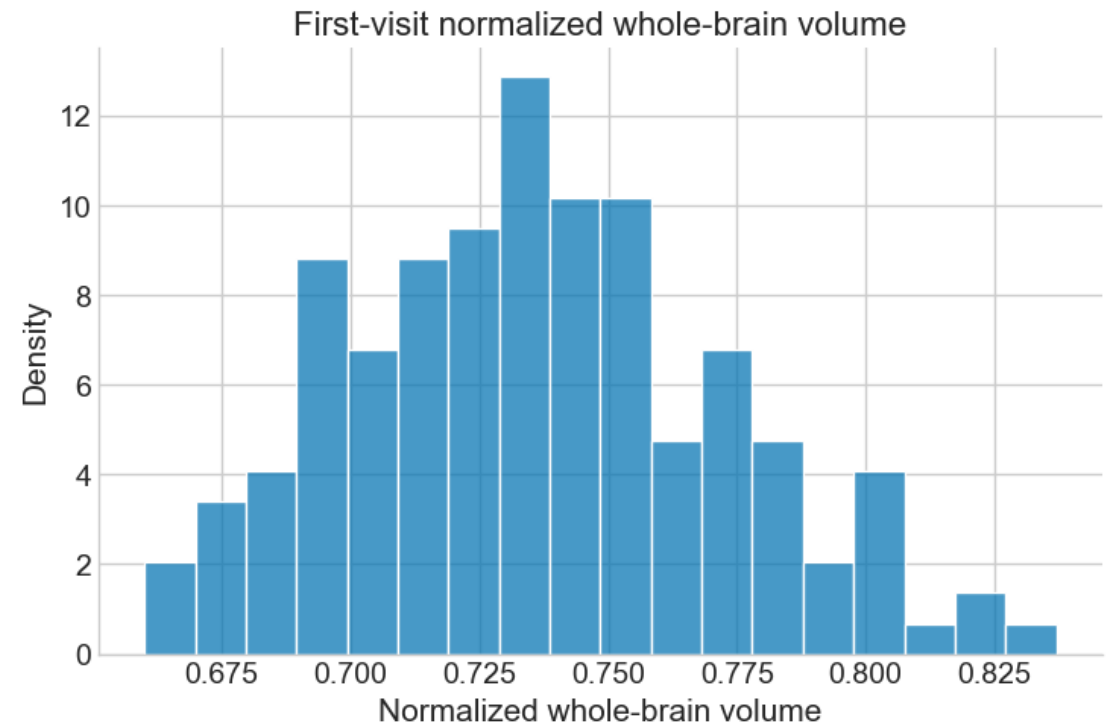
- Data recorded from 150 people (ages 60-96)
  - 373 MRI sessions (2-5 scans per individual)
  - Scans span 1-7 years per individual

| Variable | Meaning                                                 | Used for                                                  |
|----------|---------------------------------------------------------|-----------------------------------------------------------|
| nWBV     | Normalized whole-brain volume                           | Main Normal-model outcome and diagnostic-group comparison |
| eTIV     | Estimated total intracranial volume                     | Clear two-group continuous comparison by gender           |
| Group    | Diagnostic trajectory: Nondemented, Demented, Converted | Scientific grouping for nWBV comparisons                  |
| Gender   | Recorded as F or M                                      | Grouping variable for the eTIV example                    |
| MMSE     | Mini-Mental State Exam score, bounded above at 30       | Wrong-likelihood example for posterior predictive checks  |
| CDR      | Clinical Dementia Rating                                | Clinical severity context for interpreting groups         |

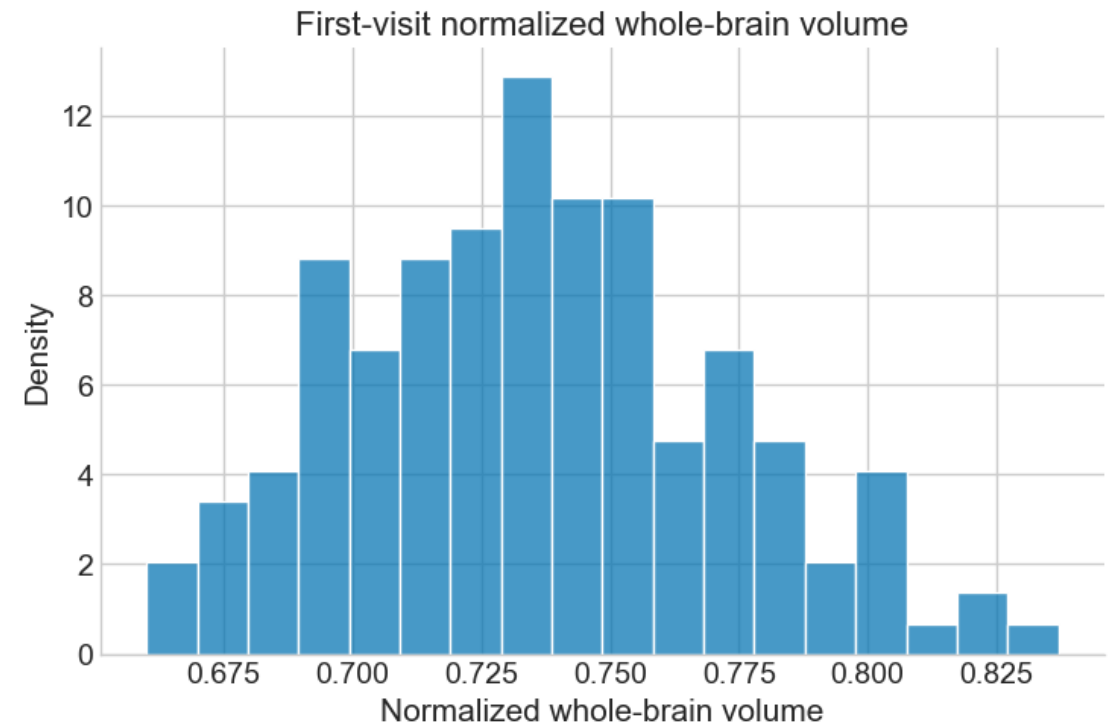
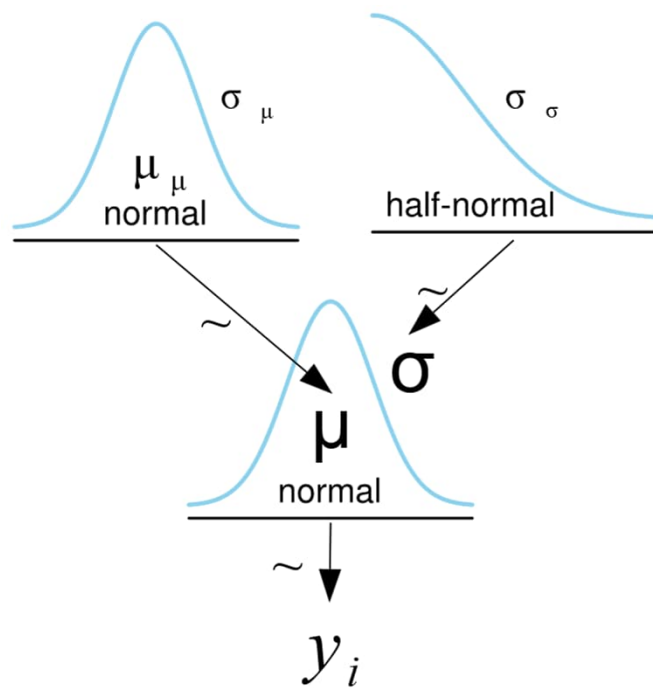
Little *IEEE Trans Biomed Eng.* 2009 Suitability of dysphonia measurements for telemonitoring of Parkinson's disease

# Continuous data is more varied than probabilities

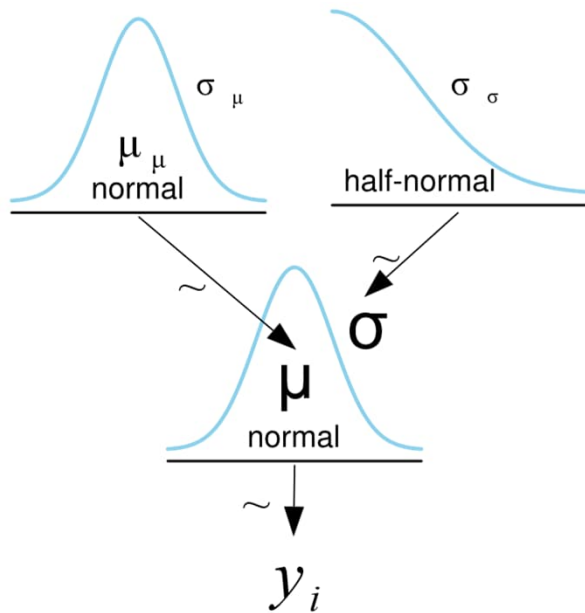
- Continuous data:
  - Comes with units
  - Characterized by a middle
    - But also a spread
  - Might be bounded
  - Might be multimodal



# Build a model in order to learn from data

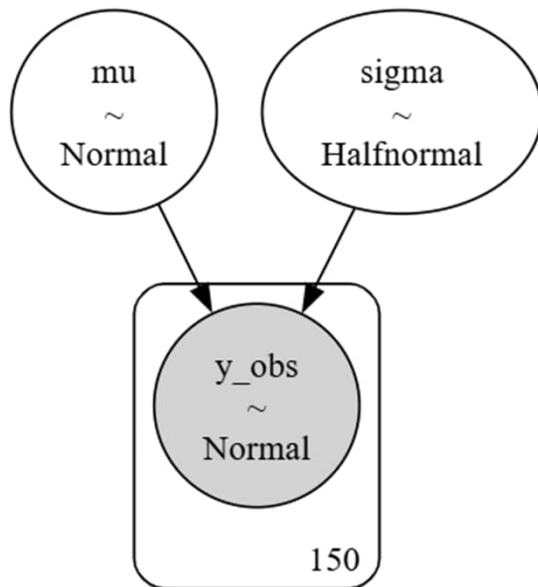


# Translate the model to PyMC



```
with pm.Model() as nwbv_normal_model:  
    mu = pm.Normal("mu", mu=mu_mu, sigma=sigma_mu)  
    sigma = pm.HalfNormal("sigma", sigma=sigma_sigma)  
    y_obs = pm.Normal("y_obs", mu=mu, sigma=sigma, observed=y)
```

# Check ourselves by translating back to graphic



```
with pm.Model() as nwbv_normal_model:  
    mu = pm.Normal("mu", mu=mu_mu, sigma=sigma_mu)  
    sigma = pm.HalfNormal("sigma", sigma=sigma_sigma)  
    y_obs = pm.Normal("y_obs", mu=mu, sigma=sigma, observed=y)
```

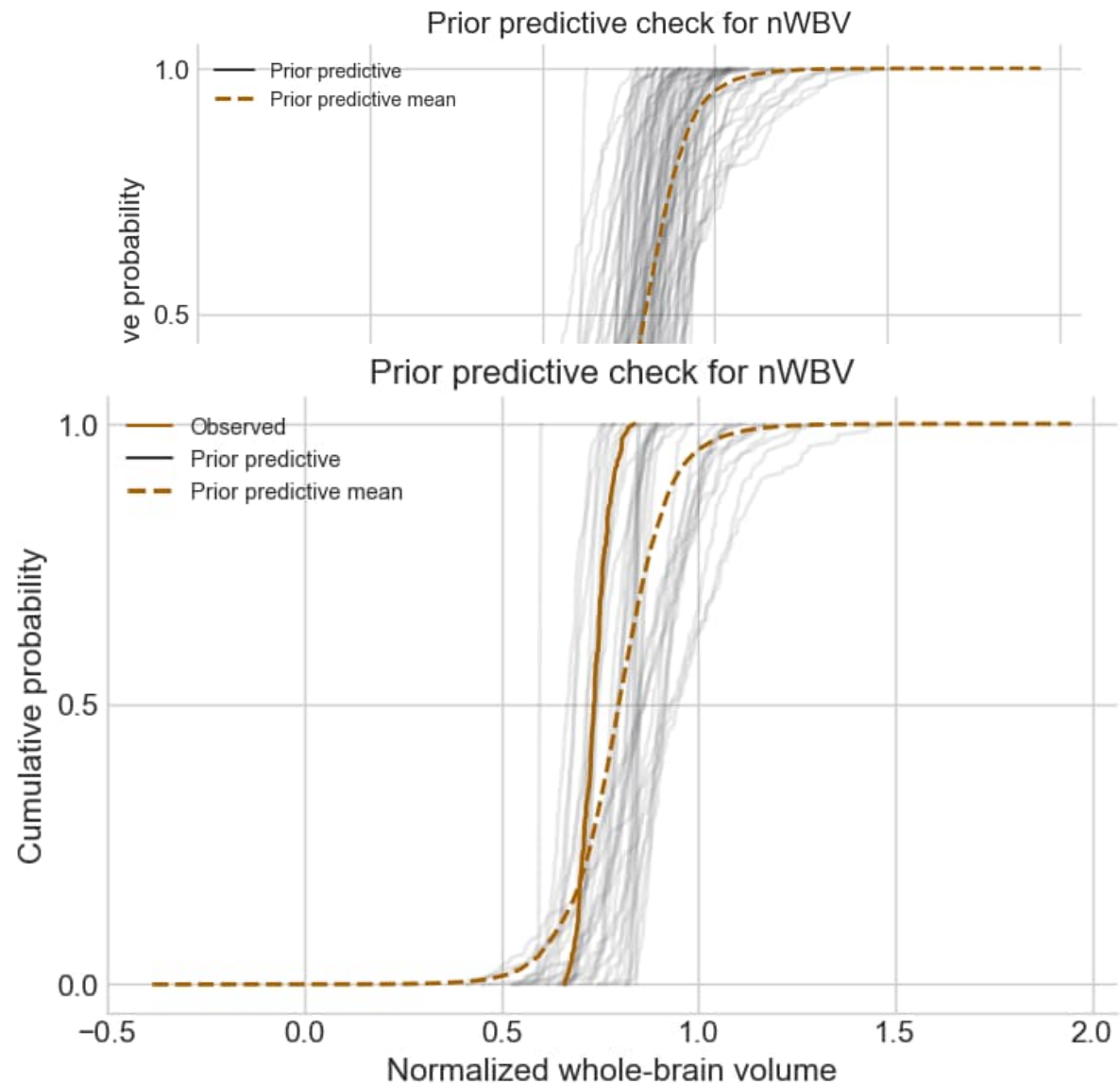


## 7.2 Normal model workflow

Prior selection, prior predictive, cumulative distributions, diagnostics, posterior predictive, Bayesian p values

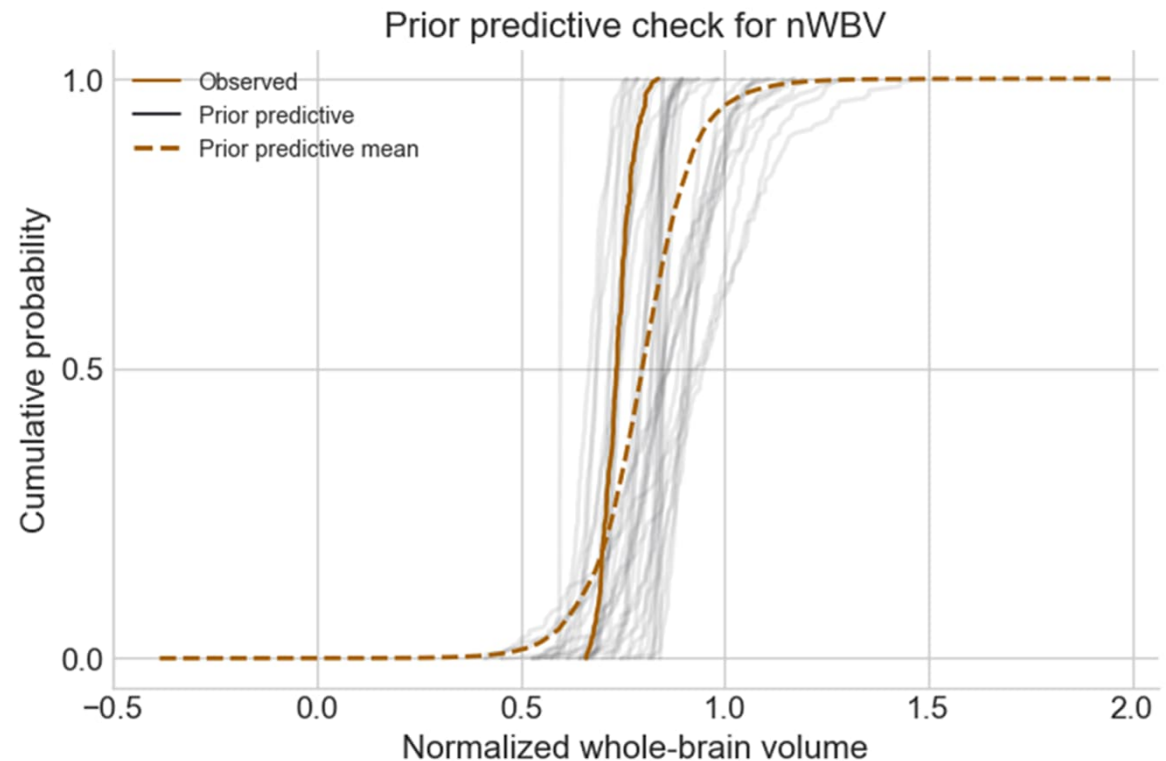
# Is prior predictive sane?

- $nWBV = WBV/tICV$ 
  - How big is the brain in the skull?
  - An expert would know!
- Where is the center?
  - More than 0.5, less than 1.0
  - $\mu_\mu = 0.8$
  - $\sigma_\mu = 0.15 \rightarrow 0.08$
- How big is the variability between people?
  - Largest range I can imagine:
    - 0.5 – 0.95
  - $\sigma_\sigma = 0.2 \rightarrow 0.10$



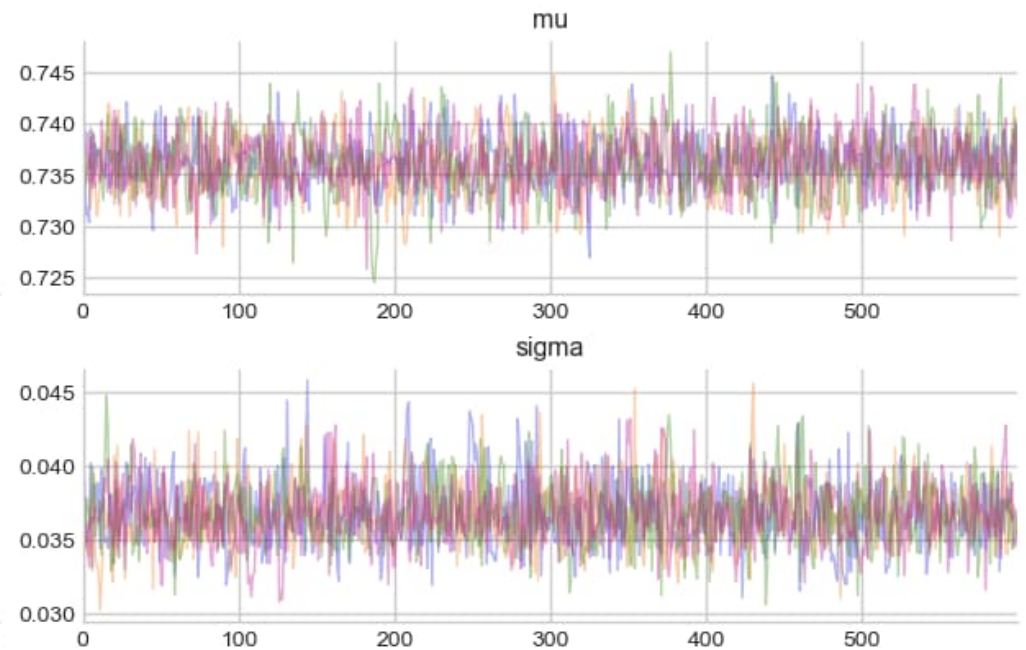
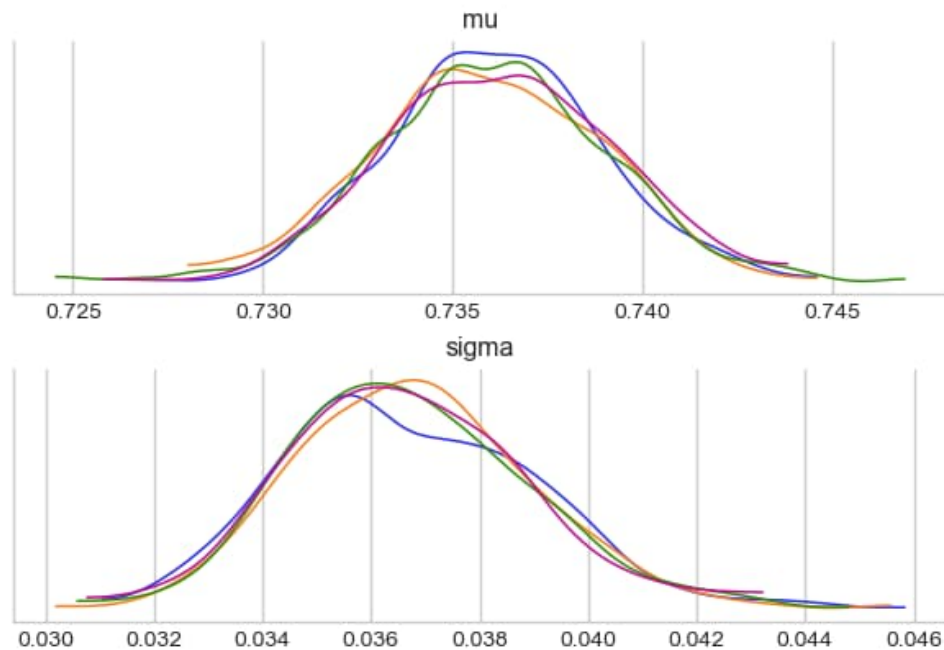
# Does our data violate our prior?

- Prior on  $\mu$ :
  - Not far from middle
- Prior on  $\sigma$ :
  - Not too steep
  - Not so flat



# Trace plots raise some questions

- Autocorrelations in trace plot
- Chain variability in posterior



# But sampling diagnostics look good

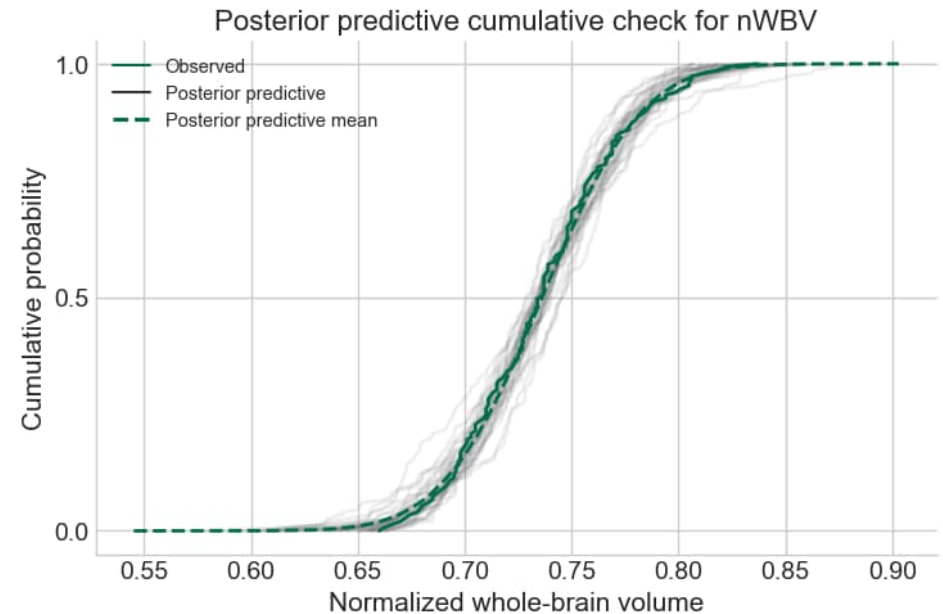
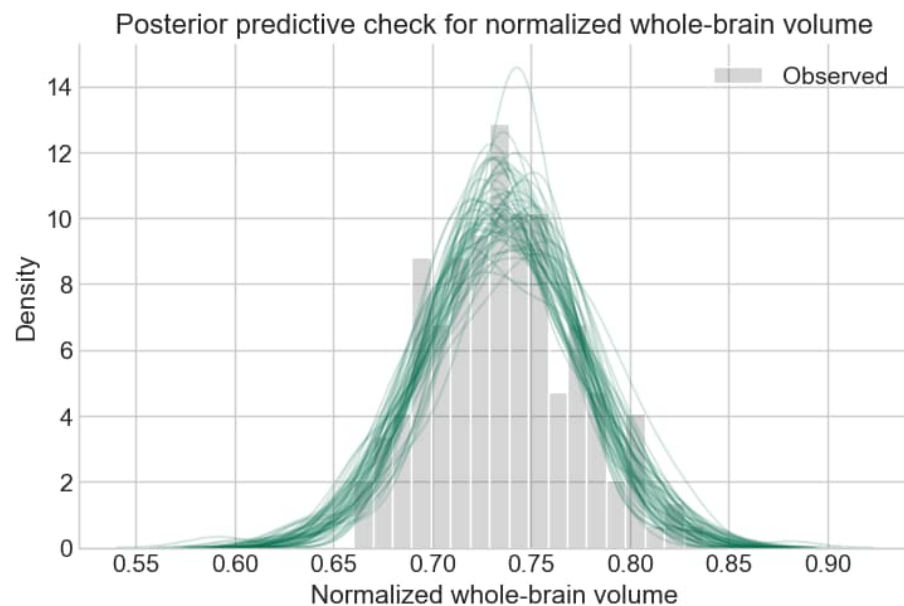
- *MCSE* is small
- *ESS* is large
- $\hat{R}$  is close to 1

|       | mean  | sd    | hdi_3% | hdi_97% | mcse_mean | mcse_sd | ess_bulk | ess_tail | r_hat |
|-------|-------|-------|--------|---------|-----------|---------|----------|----------|-------|
| mu    | 0.736 | 0.003 | 0.731  | 0.742   | 0.0       | 0.0     | 929.0    | 831.0    | 1.0   |
| sigma | 0.037 | 0.002 | 0.033  | 0.041   | 0.0       | 0.0     | 1065.0   | 569.0    | 1.0   |

For production of detailed inference, it might be worth resampling with more tuning steps, but we are probably ok.

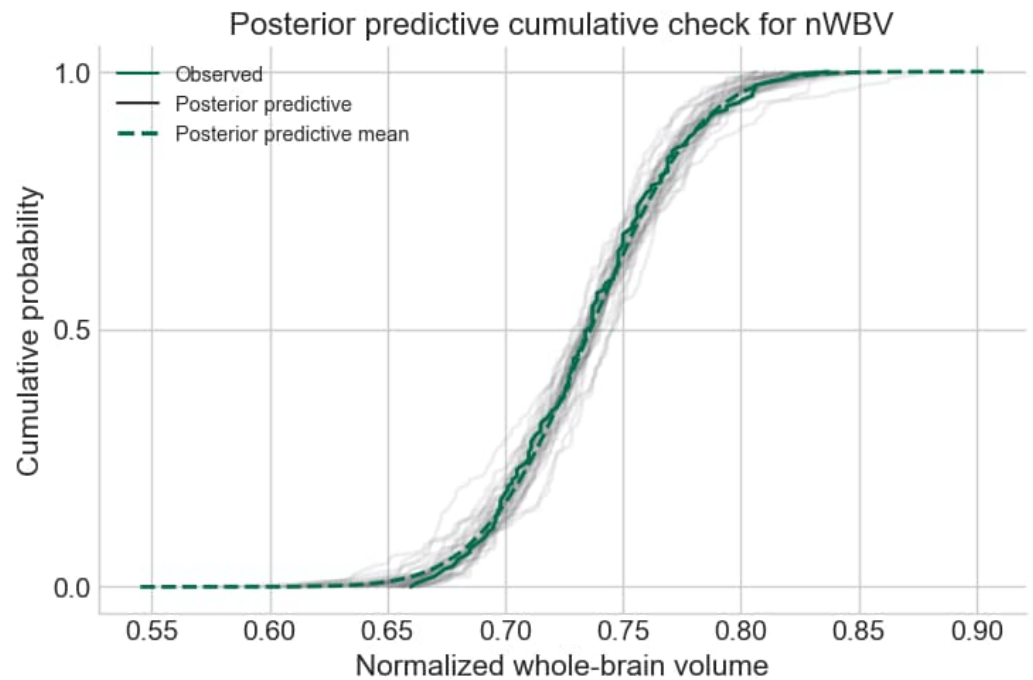
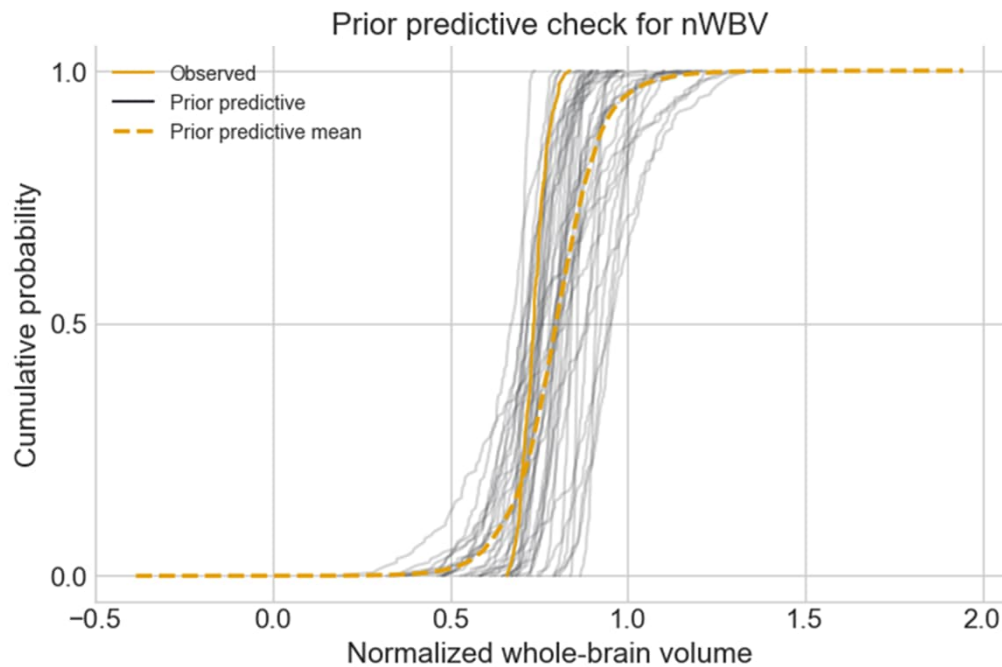
# Posterior predictive data matches real data

- Look at both posterior density and cumulative distribution



# Difference between prior and posterior predictive

- The difference shows that we've learned from our data
- Note change in x axis



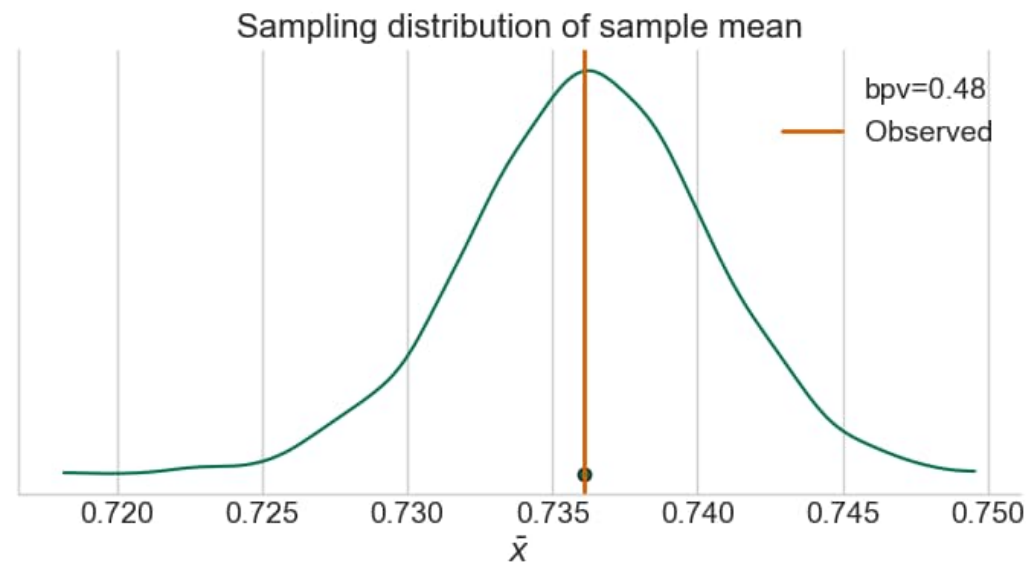
# Bayesian p values can quantify fit

- The same as frequentist p values
  - Integral of the sampling distribution more extreme than our data
  - Used as a measure of fit
- Different from frequentist p values
  - Frequentist
    - Used to show the level of significance
  - Bayesian
    - Used to show goodness of fit
  - Frequentist
    - Integrate the null sampling distribution
  - Bayesian:
    - Integrate the posterior predictive sampling distribution



# Arviz will generate a plot of the Bayesian p value

- Frequentist p values are usually calculated on known distributions
- Bayesian p values can be summed over samples instead of integrated



For each posterior sample  $\tilde{\mu}_m, \tilde{\sigma}_m$

Get a sample dataset  $\tilde{y}_{m,i} \sim N(\tilde{\mu}_m, \tilde{\sigma}_m)$

Take its sample mean  $\bar{x}_{m,i} = \sum \tilde{y}_{m,i}$

Plot the distribution of these sample means

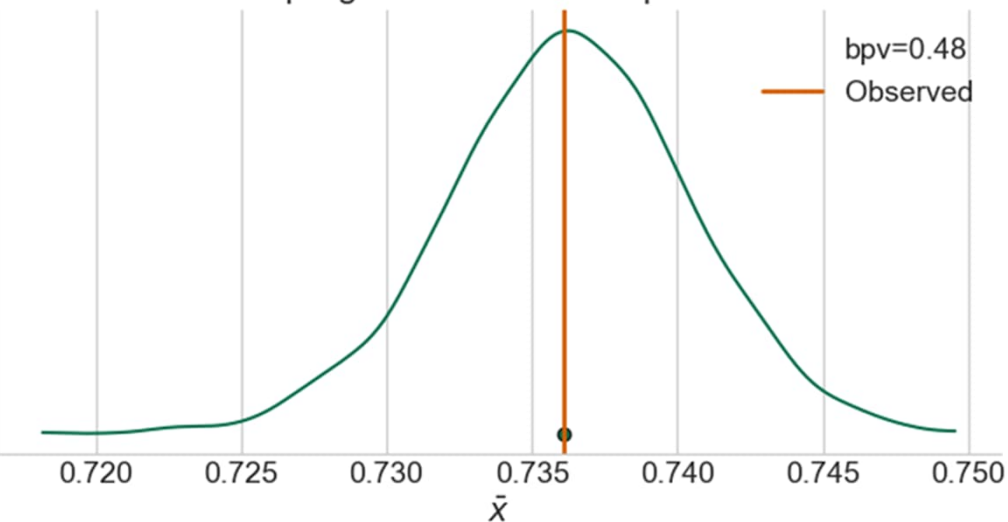
Show original data with a dot

Added a line to make it more visible

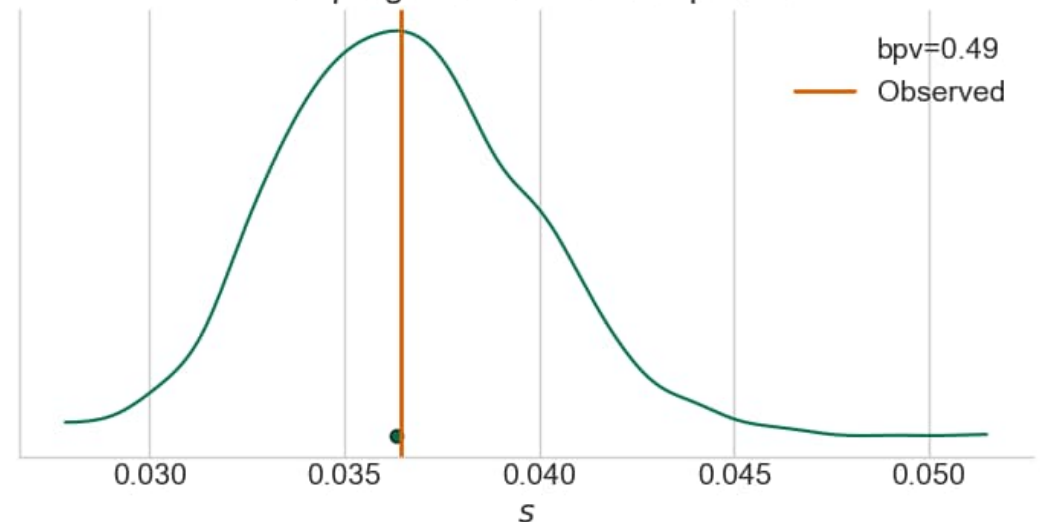
# Arviz will show Bayesian p values of different sample statistics

- The question is always:
  - How well does the fit model produce data like my data

Sampling distribution of sample mean

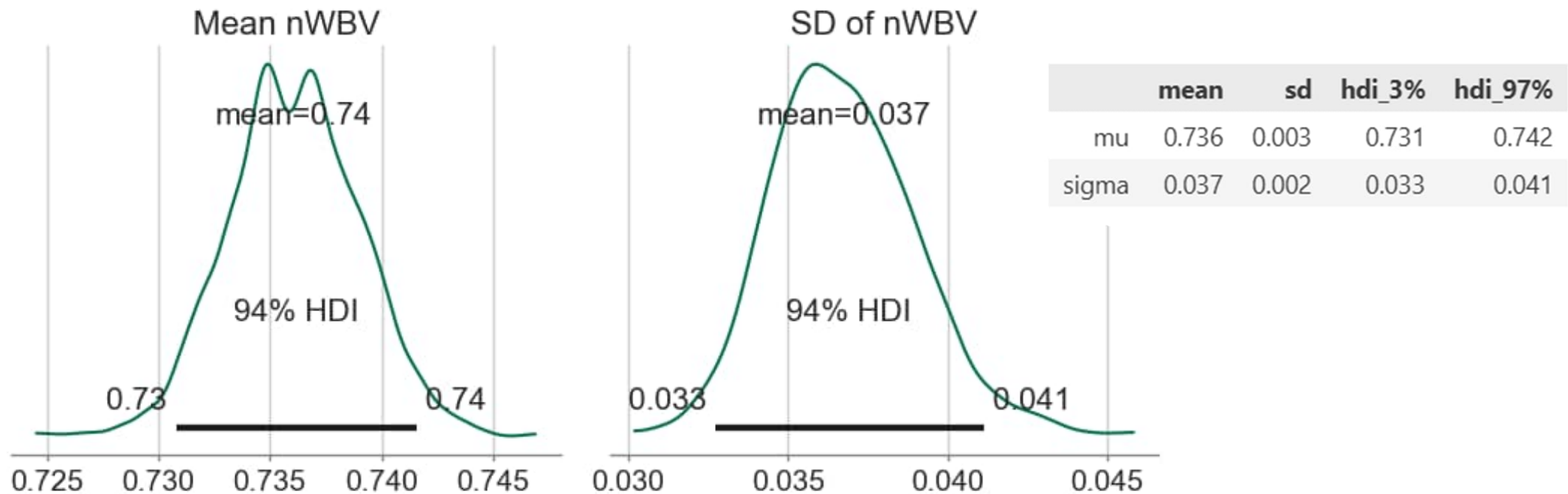


Sampling distribution of sample SD



# With workflow checks done, examine posterior

- The population average for brain size is 73.6% of tICV
- The width of the population is near 4%

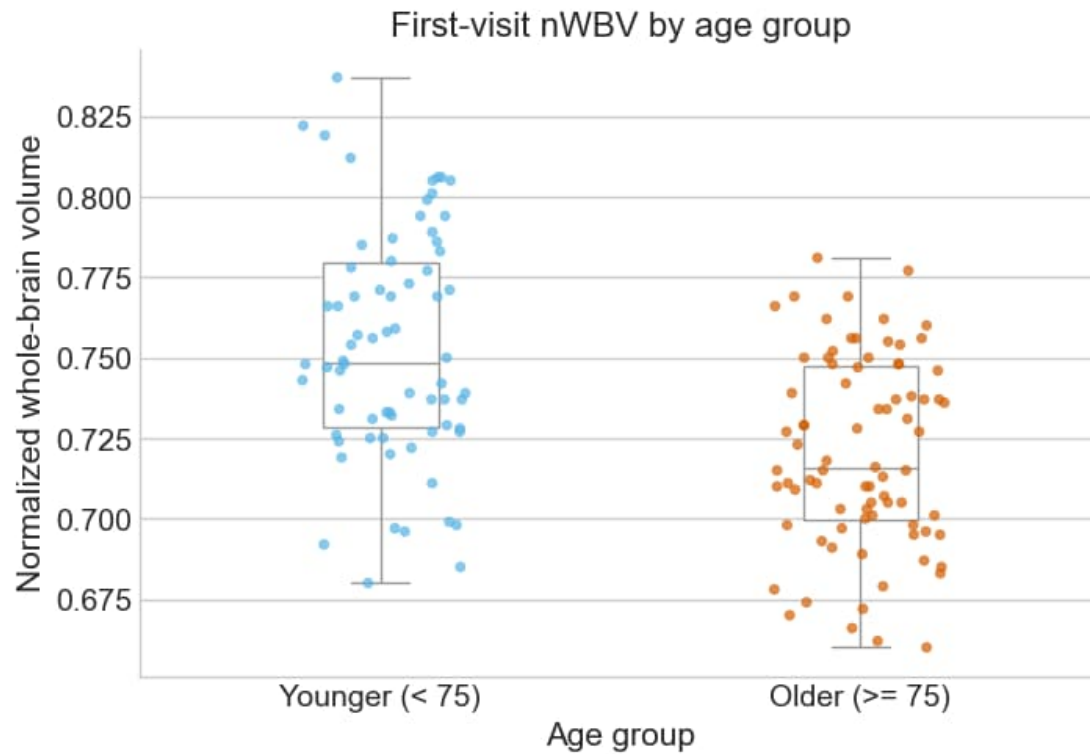


## 7.3 Comparing across ages

Comparing results from two different groups

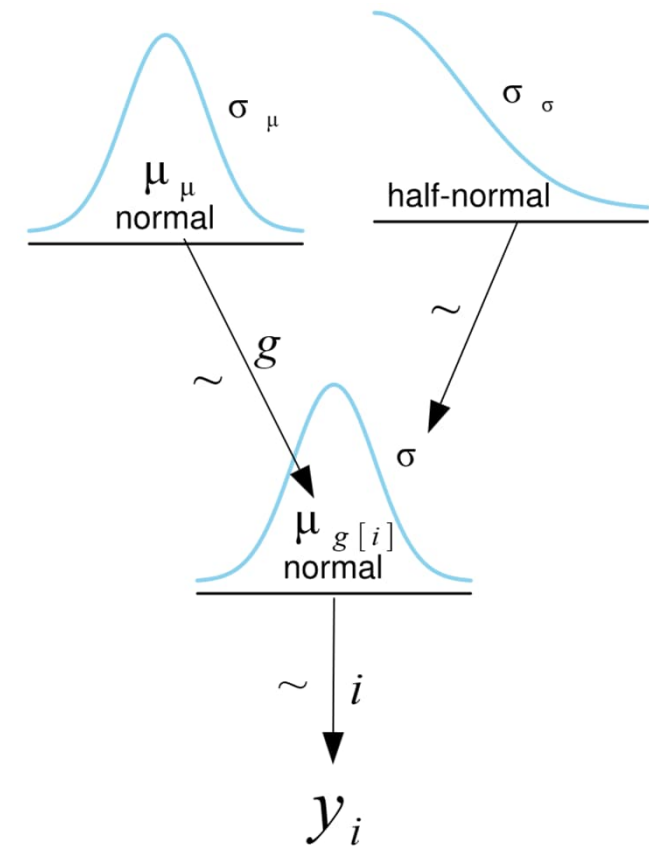
# Brain volume degenerates with age

- A well known and much studied phenomenon
- That we will use to show comparison between groups



# Model for comparing two groups

- Unpooled model
- Each group normally distributed
  - Separate mean
    - With an index selecting the right group
  - Same standard deviation
    - A common assumption
    - But not required

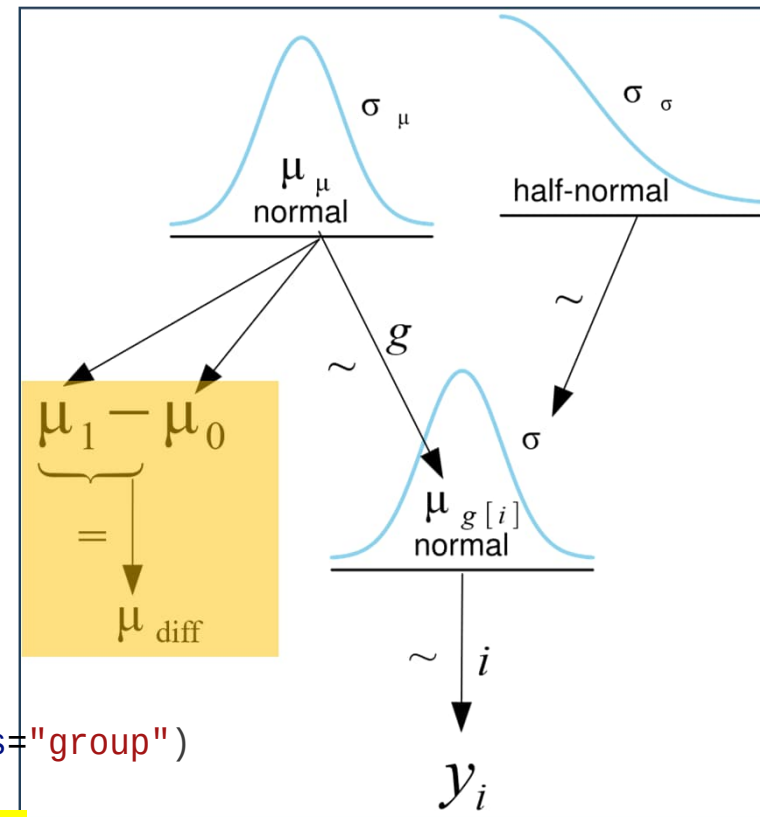


Every data point  $y_i$  has its group  $g[i]$   
Which determines the correct mean:  $\mu_{g[i]}$

# Translate model to PyMC

- Unpooled model

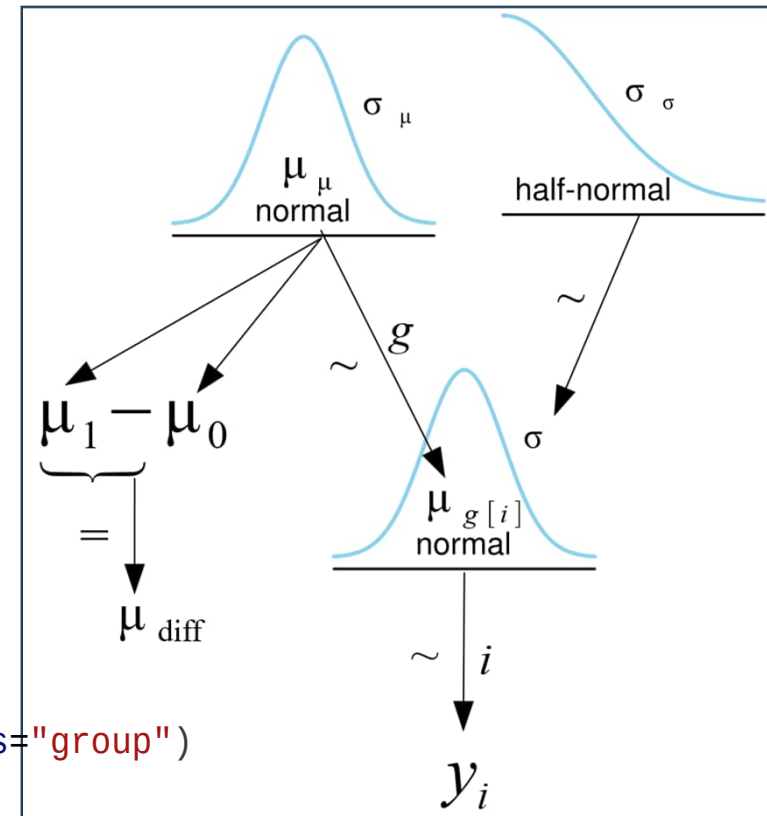
```
with pm.Model(coords=coords) as group_model:
    group = pm.Data("group", group_idx, dims="obs")
    mu_group = pm.Normal("mu_group", mu=mu_mu, sigma=sigma_mu, dims="group")
    sigma = pm.HalfNormal("sigma", sigma=sigma_sigma)
    mu_diff = pm.Deterministic("mu_diff", mu_group[1] - mu_group[0])
    nwbv_obs = pm.Normal("nwbv_obs", mu=mu_group[age_group], sigma=sigma,
                        observed=nwbv_age, dims="obs")
```



# pm.Data creates data objects

- In the idata object
- Can be updated

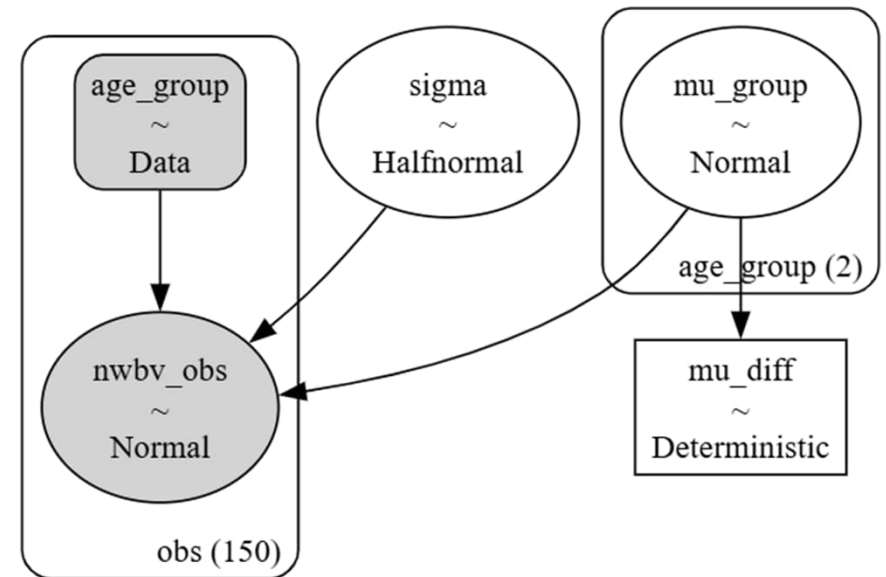
```
with pm.Model(coords=coords) as group_model:  
    group = pm.Data("group", group_idx, dims="obs")  
    mu_group = pm.Normal("mu_group", mu=mu_mu, sigma=sigma_mu, dims="group")  
    sigma = pm.HalfNormal("sigma", sigma=sigma_sigma)  
    mu_diff = pm.Deterministic("mu_diff", mu_group[1] - mu_group[0])  
    nwbv_obs = pm.Normal("nwbv_obs", mu=mu_group[age_group], sigma=sigma,  
                        observed=nwbv_age, dims="obs")
```





# Graphic visualization verifies model

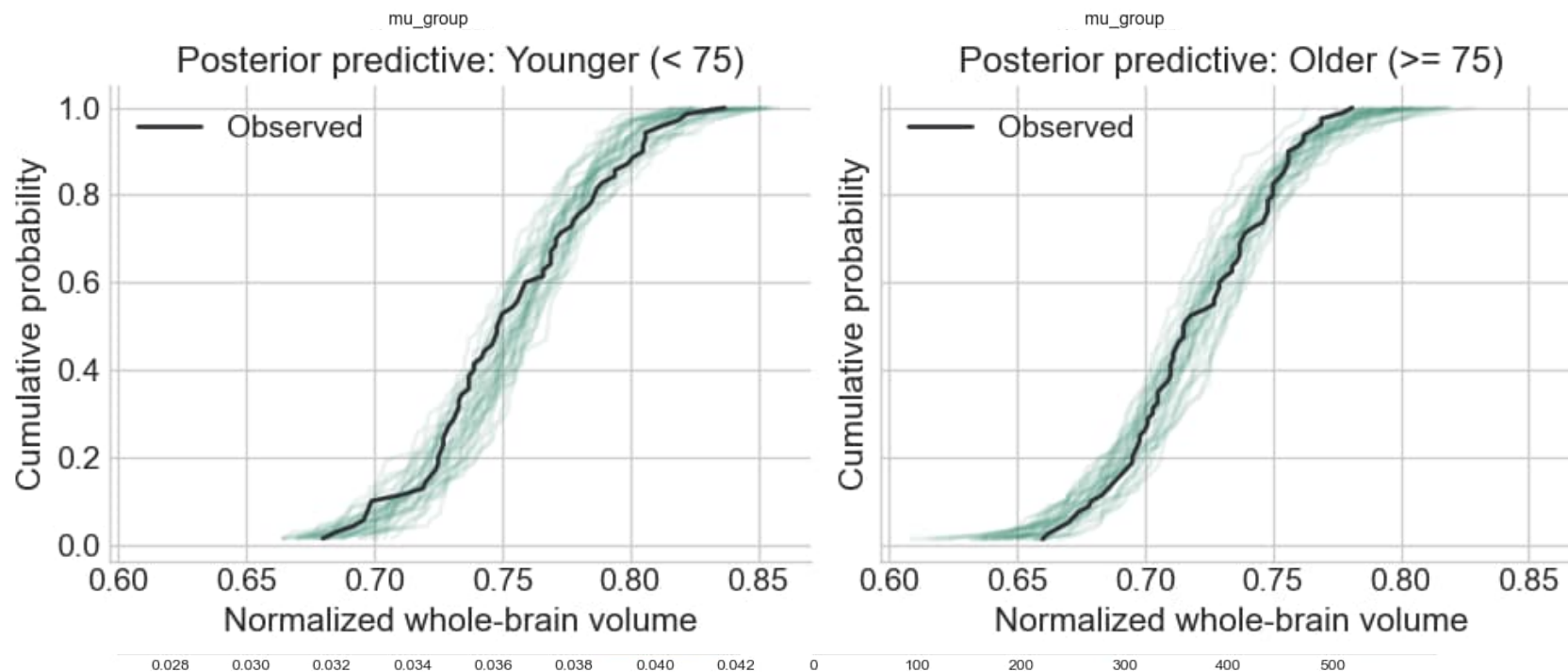
- Grey for observed or known
- Square for constant or deterministic
- Box for multidimensional



```
with pm.Model(coords=coords) as group_model:
    group = pm.Data("age_group", group_idx, dims="obs")
    mu_group = pm.Normal("mu_group", mu=mu_mu, sigma=sigma_mu, dims="group")
    sigma = pm.HalfNormal("sigma", sigma=sigma_sigma)
    mu_diff = pm.Deterministic("mu_diff", mu_group[1] - mu_group[0])
    nbwv_obs = pm.Normal("nwbv_obs", mu=mu_group[age_group], sigma=sigma,
                        observed=nwbv_age, dims="obs")
```

# Bayesian workflow

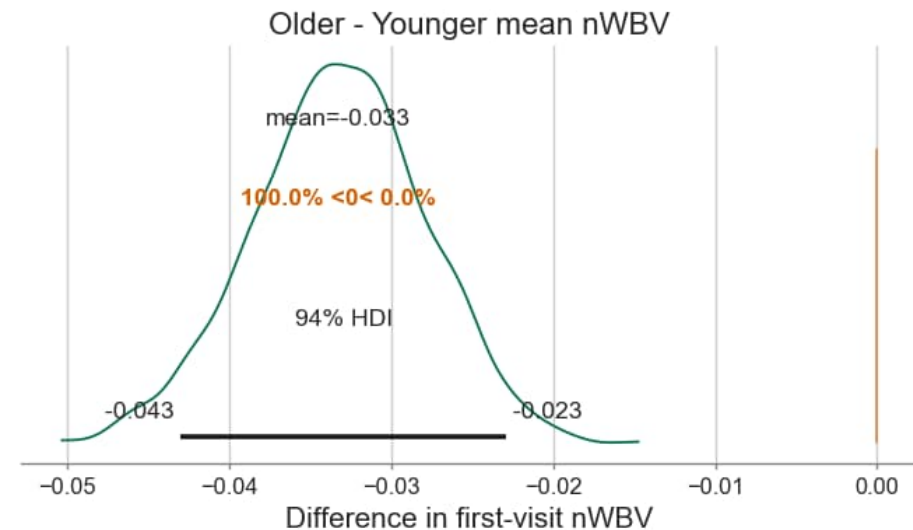
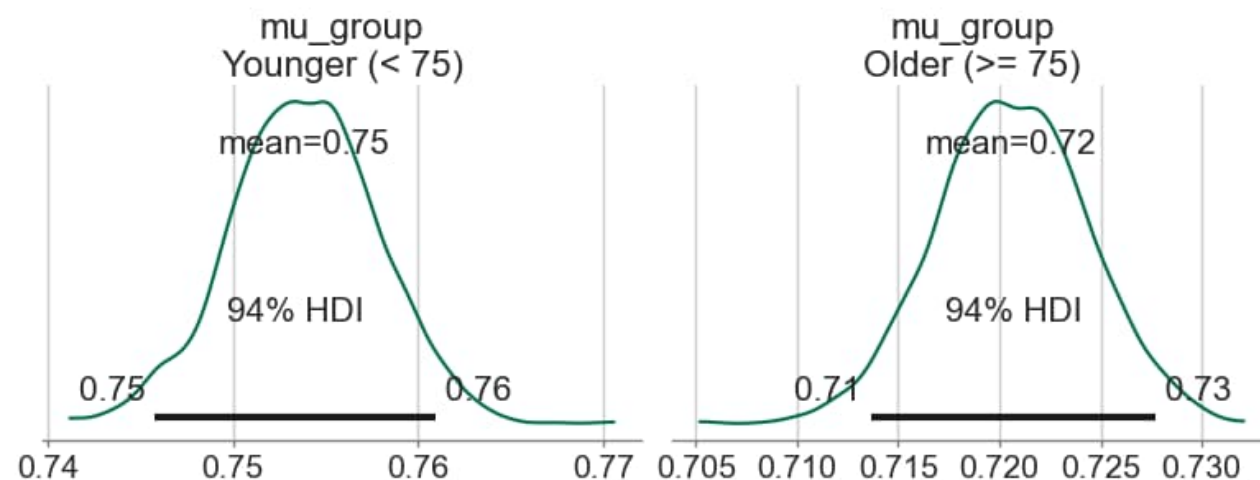
- Prior predictive, diagnostics, posterior predictive
  - These actually required model tuning while preparing the lecture



# Examine posteriors

- We care about the means
- And about the difference

Posterior mean nWBV by age group



## 7.4 Effect sizes for normal models

Raw effect sizes, Cohen's  $d$ , HDI/ROPE hypothesis testing,  $t$  tests

# Effect sizes for the difference between two groups

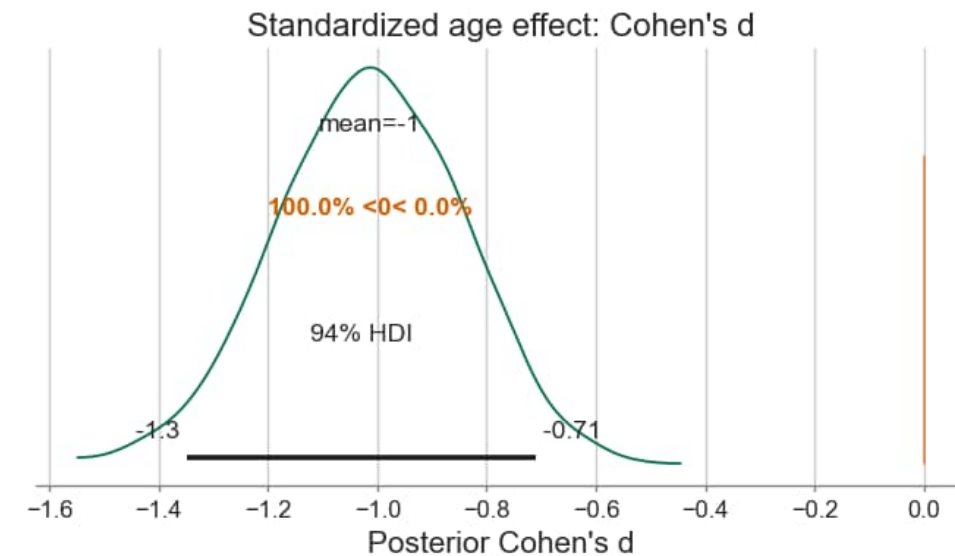
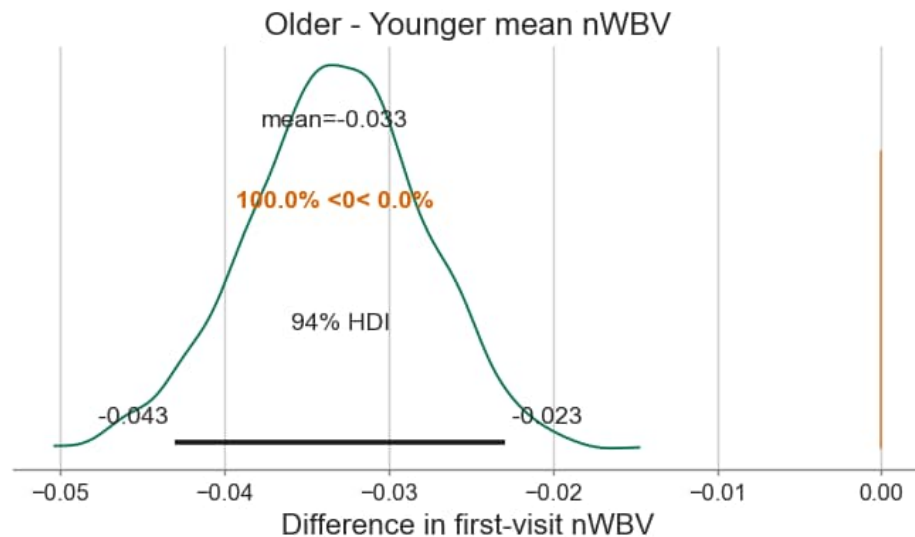
- Raw effect size
  - The difference in measured units
    - Present tICV
- Normalized effect size
  - Cohen's d
    - Assumes a single variance in the denominator
    - If we had modeled two variances
      - The formula would be different

$$\text{Cohen's } d = \frac{\mu_1 - \mu_2}{\sigma}$$

$$\text{or } \frac{\mu_1 - \mu_2}{\sqrt{\frac{\sigma_1^2 + \sigma_2^2}{2}}}$$

| Effect size | Cohen's d |
|-------------|-----------|
| None        | 0         |
| Small       | 0.2       |
| Medium      | 0.5       |
| Large       | 0.8       |

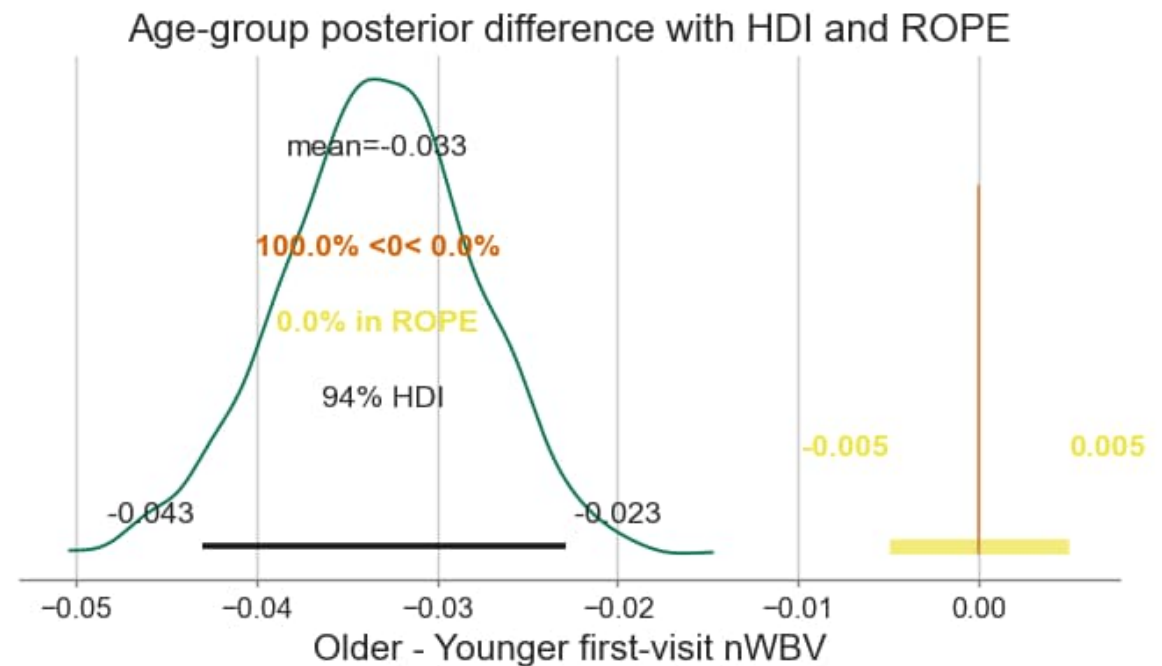
# Both raw and normalized effect sizes have distributions



```
with pm.Model(coords=coords) as group_model:
    group = pm.Data("group", group_idx, dims="obs")
    mu_group = pm.Normal("mu_group", mu=mu_mu, sigma=sigma_mu, dims="group")
    sigma = pm.HalfNormal("sigma", sigma=sigma_sigma)
    mu_diff = pm.Deterministic("mu_diff", mu_group[1] - mu_group[0])
    cohen_d = pm.Deterministic("cohen_d", mu_diff / sigma)
    nwbv_obs = pm.Normal("nwbv_obs", mu=mu_group[age_group], sigma=sigma, observed=nwbv_age, dims="obs")
```

# HDI / ROPE hypothesis testing for normal models

- Arbitrarily chose ROPE to be  $\pm 0.5\%$  tICV
- Given that most brains are over 65% tICV
  - 0.5% doesn't seem very big



# Frequentist Cohen's d is a function of the data

- Frequentist statistics don't have distributions
  - They are calculated on the data

$$\text{Frequentist Cohen's } d = \frac{\bar{y}_1 - \bar{y}_2}{s}$$

$$\text{Sample mean } \bar{y}_k = \frac{1}{N_k} \sum_{i=1}^{N_k} y_{k,i}$$

$$\text{Pooled estimate of the mean } s = \sqrt{\frac{(N_1 - 1)s_1^2 + (N_2 - 1)s_2^2}{N_1 + N_2 - 2}}$$

$$\text{Sample variance } s_k^2 = \frac{1}{N_k - 1} \sum_{i=1}^{N_k} (y_{k,i} - \bar{y}_k)^2$$

$$\text{Sometimes also } s = \sqrt{\frac{(N_1 - 1)s_1^2 + (N_2 - 1)s_2^2}{N_1 + N_2}}$$

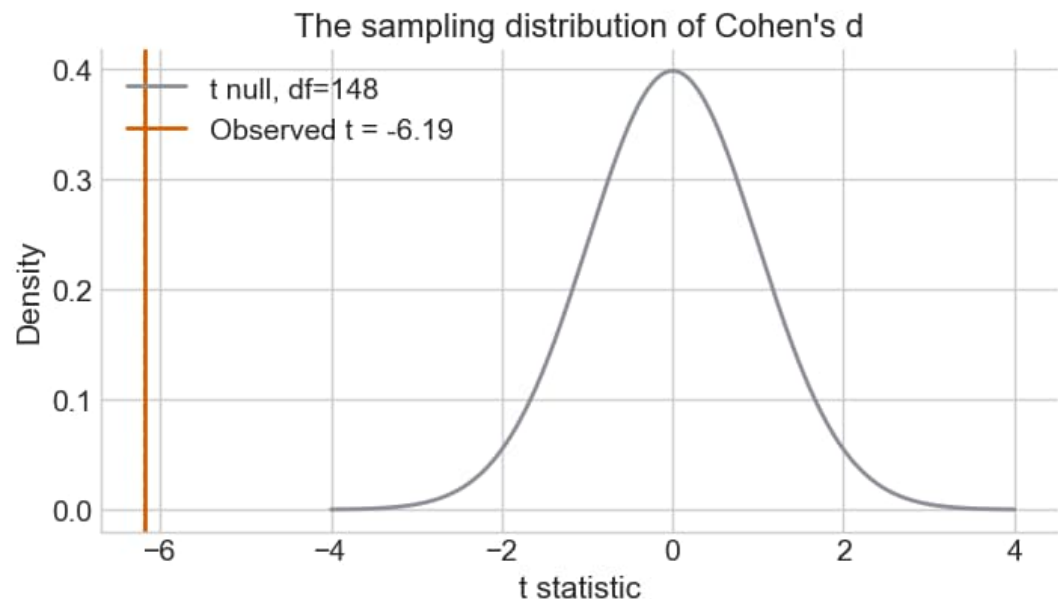


# Frequentist Cohen's d has a sampling distribution

- Frequentist statistics don't have distributions
  - But they do have sampling distributions under specific models
    - Especially the null model
  - Cohen's d has a t distribution
    - With  $N_1 + N_2 - 2$  degrees of freedom

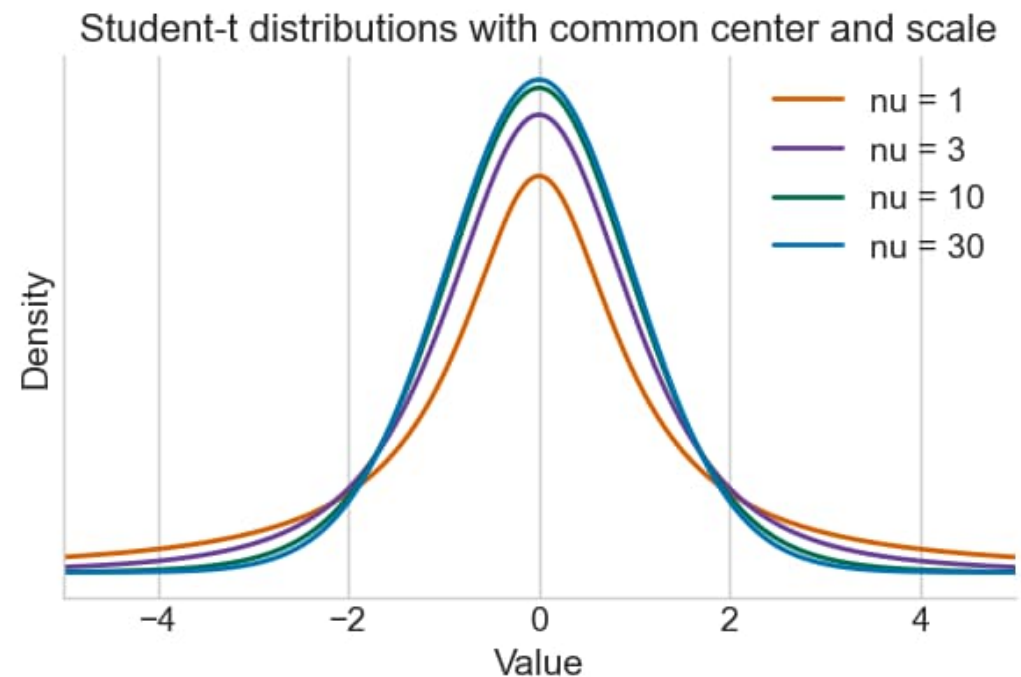
$$\text{Cohen's } d = \frac{\bar{y}_1 - \bar{y}_2}{s}$$

Cohen's  $d \sim \text{Student's } t(\mu = 0, \sigma = 1, \nu = N_1 + N_2 - 2)$



# The Student's t distribution

- A distribution with 3 parameters
  - $\mu$  : The center
    - Called the mean
    - Equals the mean
  - $\sigma$  : The width
    - Called the standard deviation or the shape
    - Not equal to the standard deviation
  - $\nu$  : The kurtosis
    - Called the degrees of freedom
    - Or the normality parameter
    - When  $\nu \rightarrow \infty$ 
      - Student's t  $\rightarrow$  Normal
    - When  $\nu \rightarrow 0$ 
      - Student's t  $\rightarrow$  Very large kurtosis

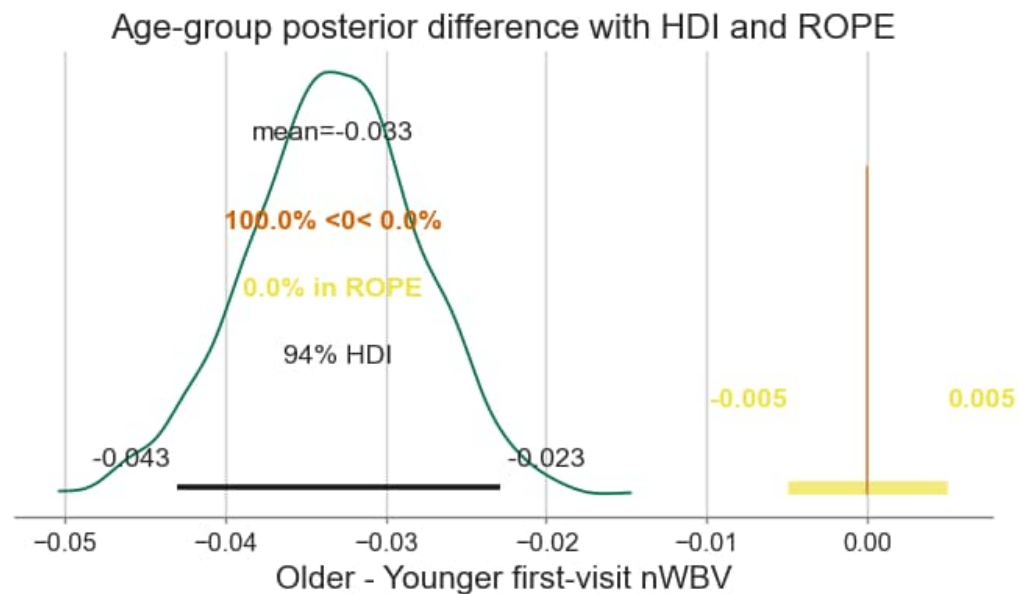


# A frequentist test for comparing two groups

- The t test
  - Compares frequentist Cohen's d
    - To the t distribution
    - With the correct degrees of freedom

| Statistic                   | Value      |
|-----------------------------|------------|
| Younger mean                | 0.753686   |
| Older mean                  | 0.720662   |
| Younger SD                  | 0.035784   |
| Older SD                    | 0.029551   |
| Pooled SD                   | 0.032605   |
| Cohen's d (Older - Younger) | -1.012819  |
| Degrees of freedom          | 148.000000 |
| 94% two-sided  t  threshold | 1.895319   |
| Two-sided p-value           | 0.000000   |

## Bayesian hypothesis test for comparison

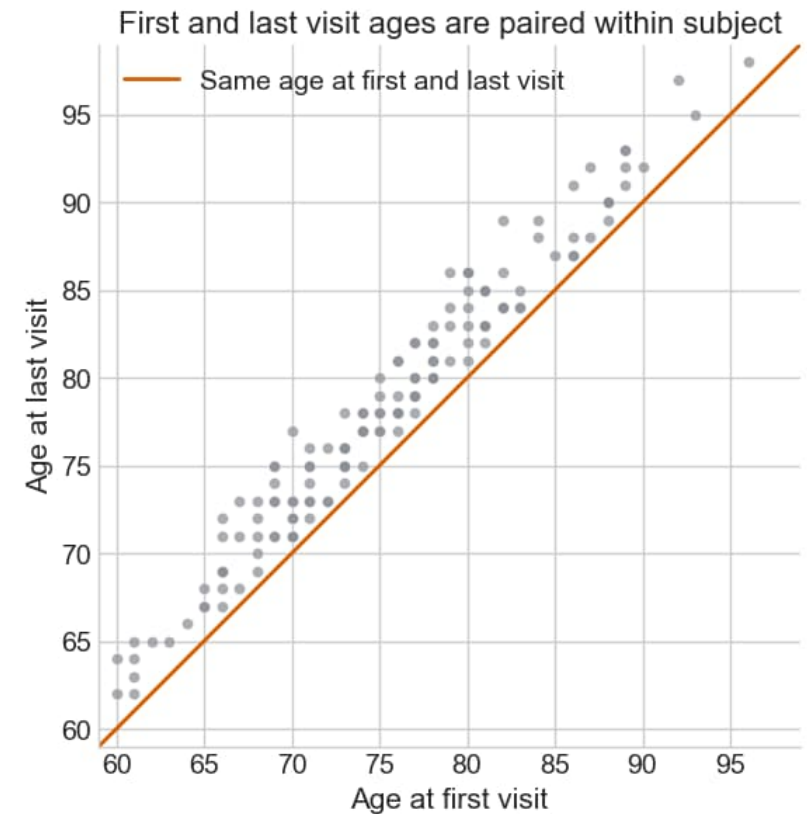
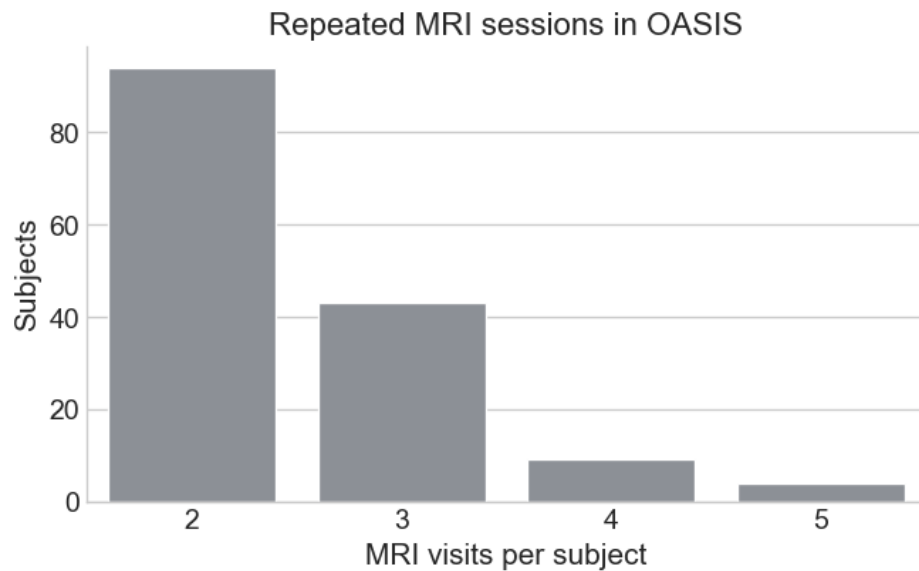


## 7.5 Paired comparisons

Comparing across groups is different than comparing within groups

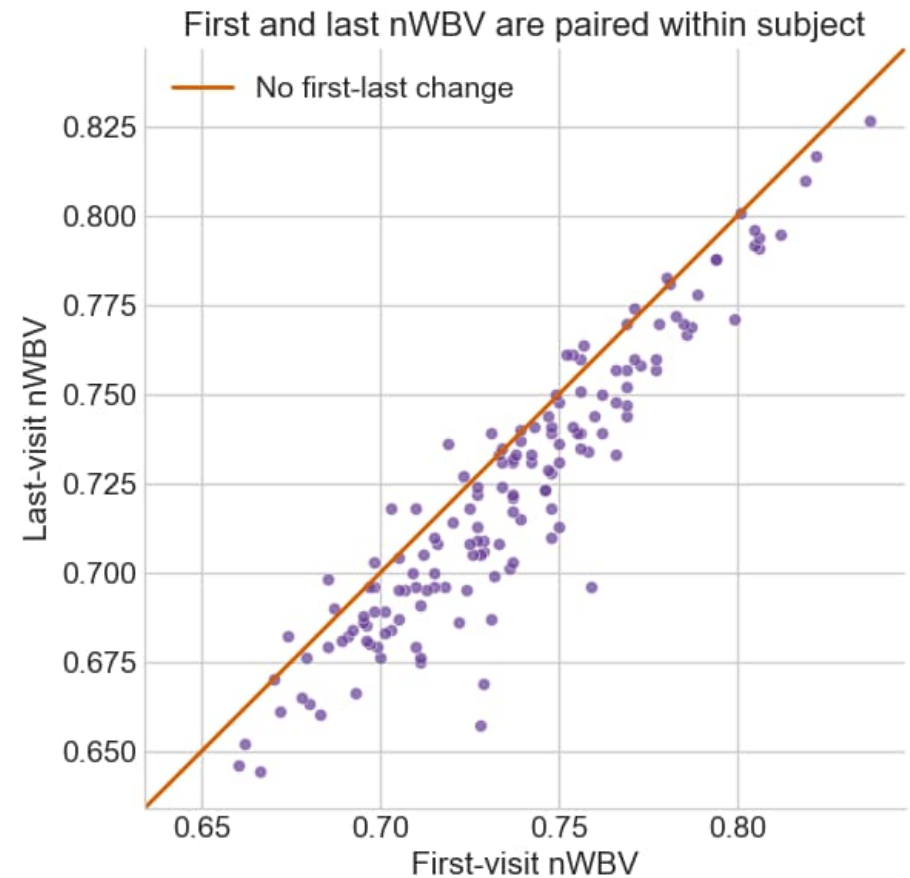
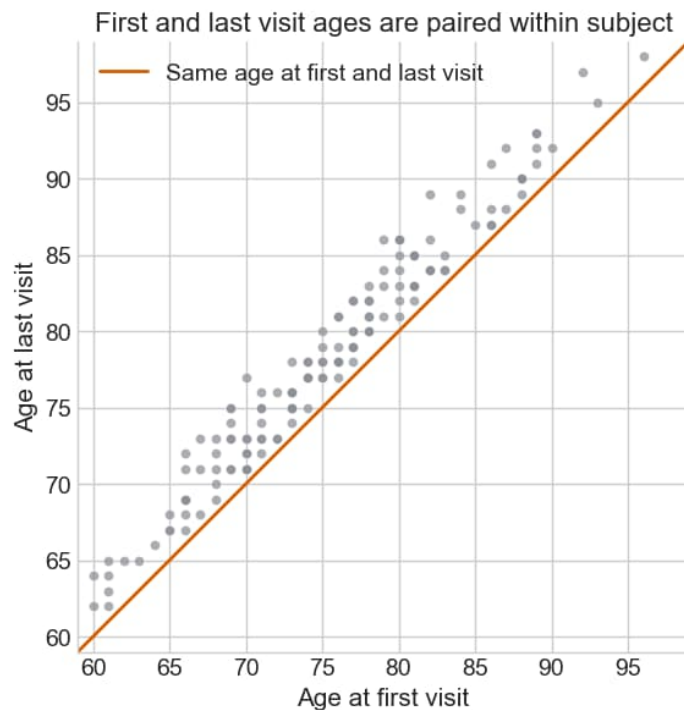
# Each participant did a number of MRI scans

- Visits were separated by 1-5 years



# Variability between subjects can hide consistency

- Almost all subjects
  - Had smaller normalized brain volume
  - On the last visit



# Paired and unpaired measures of effect of aging

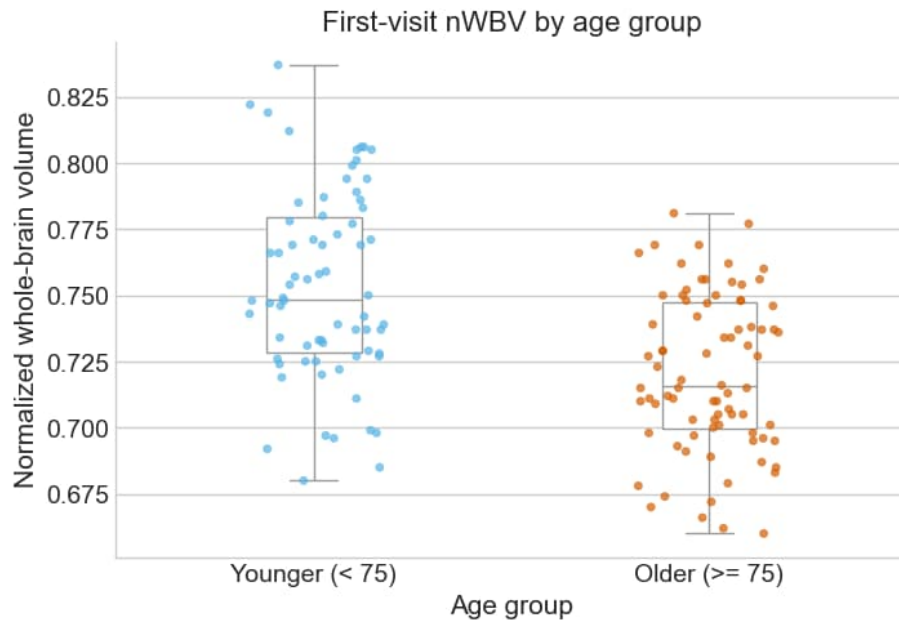
- Unpaired data:

- Often simpler to collect
- Safe from order effects

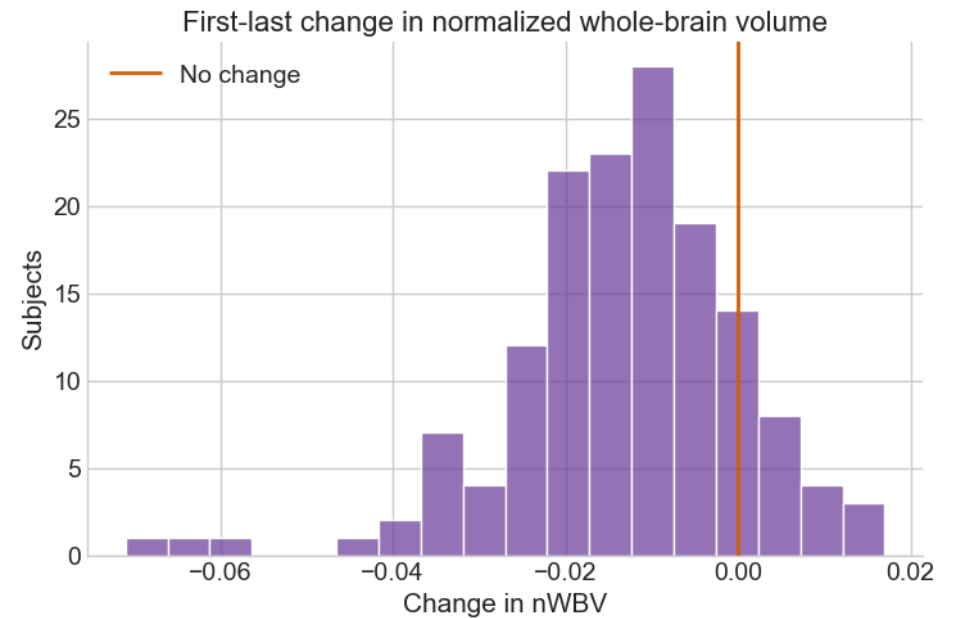
- Paired data:

- Reduces effects of between subject variability
- Number of subjects required

Unpaired: difference in nWBV across ages



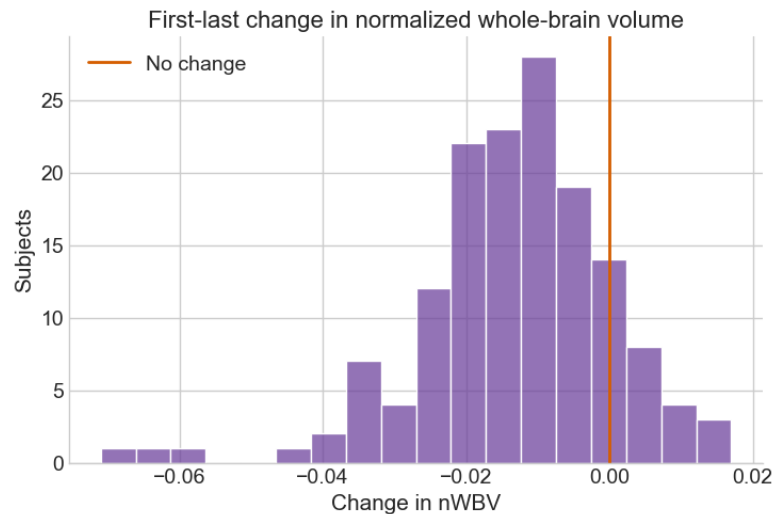
Paired: reduction in nWBV for each subject



# The paired model is just the normal model

- Except that our prior for  $\mu$  will be around 0

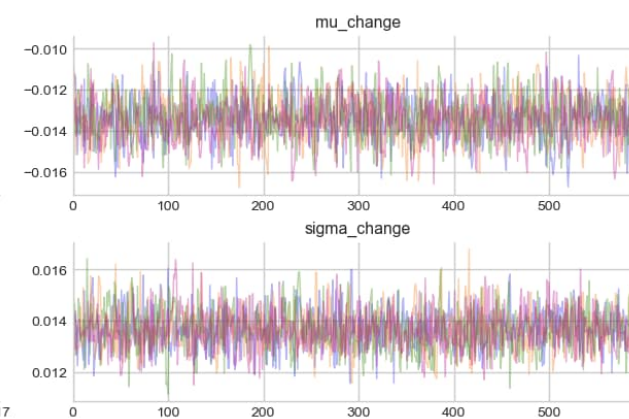
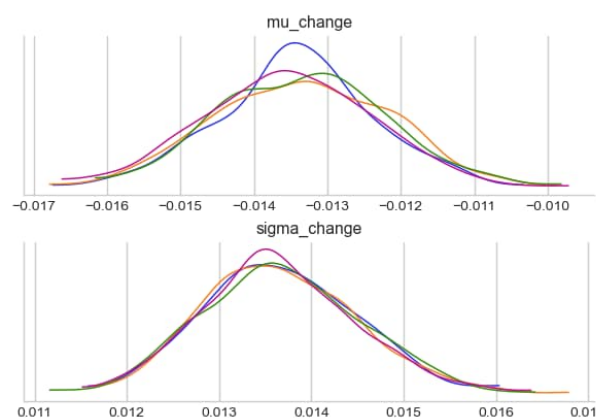
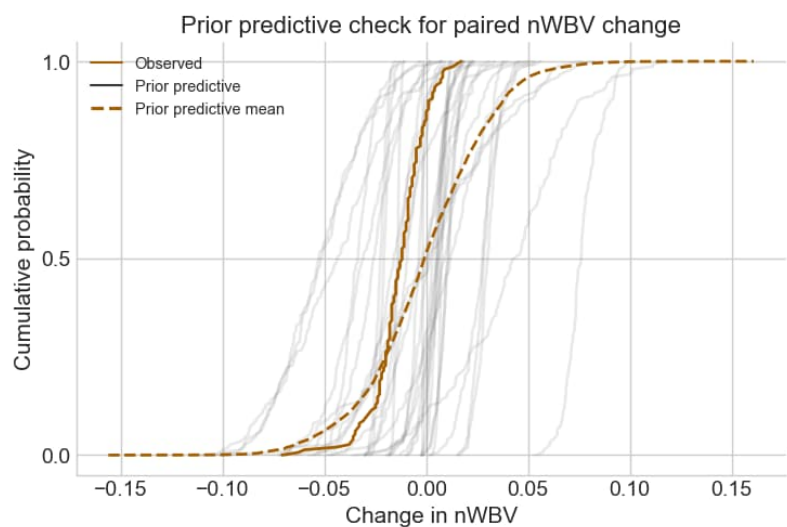
```
with pm.Model() as paired_normal_model:  
    mu_change = pm.Normal("mu_change", mu=mu_mu_change, sigma=sigma_mu_change)  
    sigma_change = pm.HalfNormal("sigma_change", sigma=sigma_sigma_change)  
    change_obs = pm.Normal("change_obs", mu=mu_change, sigma=sigma_change, observed=change_paired)
```





# Apply the Bayesian workflow

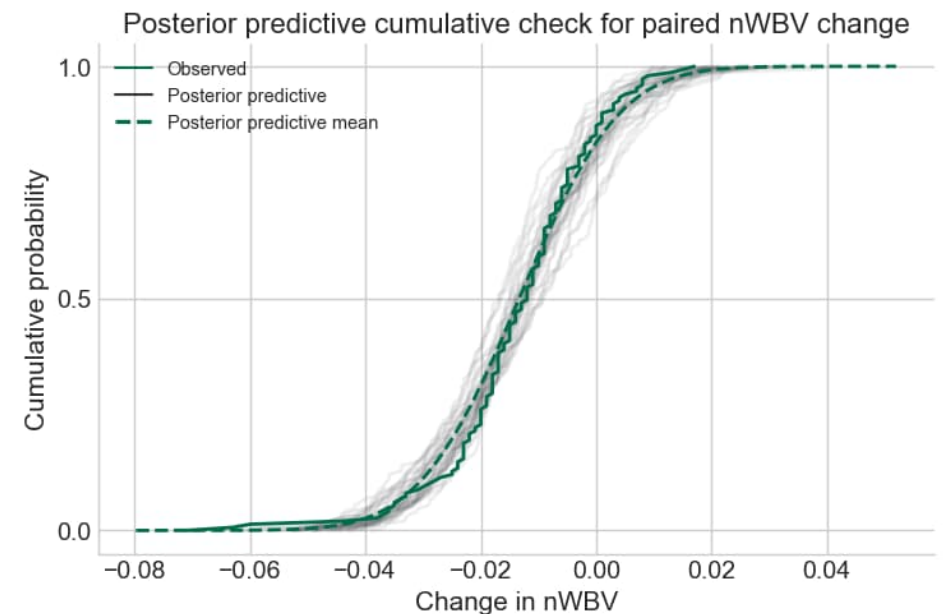
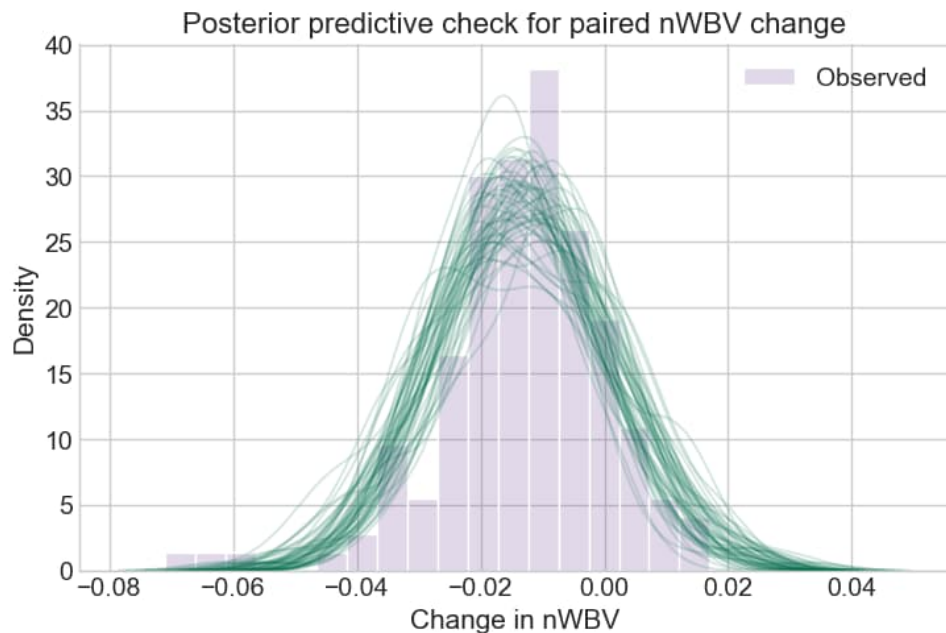
- Prior and trace do not seem unreasonable



|              | mean   | sd    | hdi_3% | hdi_97% | mcse_mean | mcse_sd | ess_bulk | ess_tail | r_hat |
|--------------|--------|-------|--------|---------|-----------|---------|----------|----------|-------|
| mu_change    | -0.013 | 0.001 | -0.015 | -0.011  | 0.0       | 0.0     | 1745.0   | 1410.0   | 1.0   |
| sigma_change | 0.014  | 0.001 | 0.012  | 0.015   | 0.0       | 0.0     | 2336.0   | 1617.0   | 1.0   |

# Posterior check raises concerns

- Some data are poorly explained by the model
  - Outliers
    - Normal models are sensitive to outliers
  - We need a likelihood that is less sensitive to outliers!

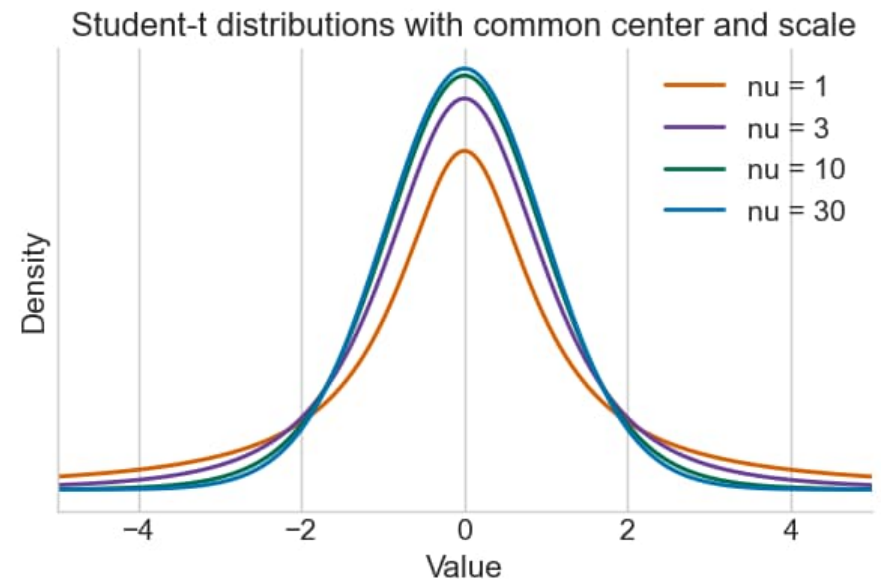


## 7.6 Robust models

The student's  $t$  distribution, the degrees of freedom parameter

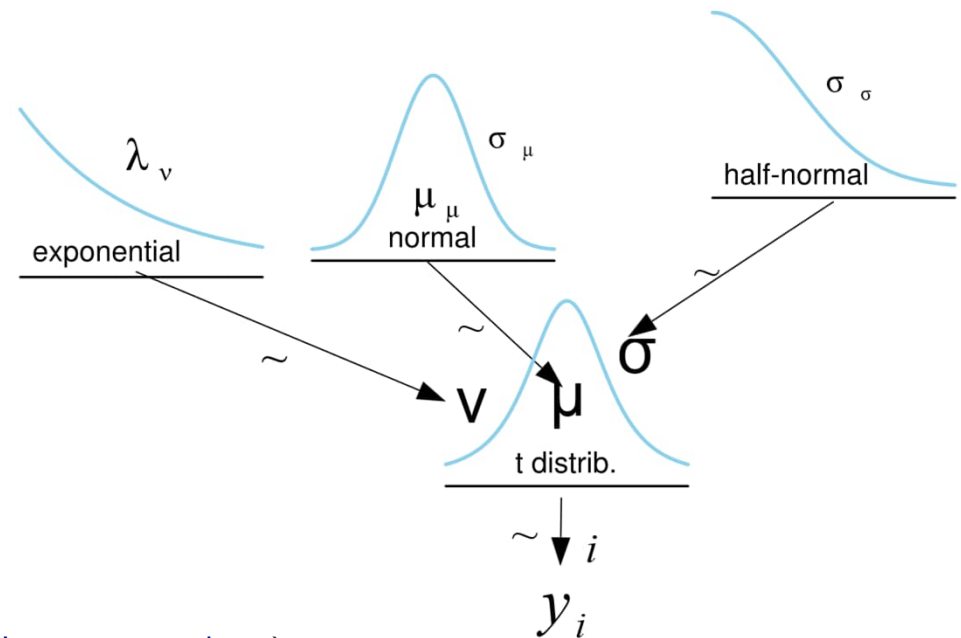
# We need a likelihood that is less sensitive to outliers

- Some data are poorly explained by the model
  - Normal models are sensitive to outliers
  - We need a likelihood that is less sensitive to outliers
- We have just learned about one!
  - The t distribution has thick tails
  - Which allows the possibility of some outliers



# As Bayesians, we can choose our model

- Replace normal likelihood with Student's t likelihood
- One more parameter
  - The 'normality' parameter
    - If posterior is small : robust to outliers
    - If posterior is large : converge to normal model



```
with pm.Model() as paired_student_t_model:  
    mu_change = pm.Normal("mu_change", mu=mu_mu_student, sigma=sigma_mu_student)  
    sigma_change = pm.HalfNormal("sigma_change", sigma=sigma_sigma_student)  
    nu = pm.Exponential("nu", lam=lam_nu_student)  
    change_obs = pm.StudentT("change_obs", nu=nu, mu=mu_change, sigma=sigma_change, observed=change_paired)
```

# It is safer to stop $\nu$ from becoming too small

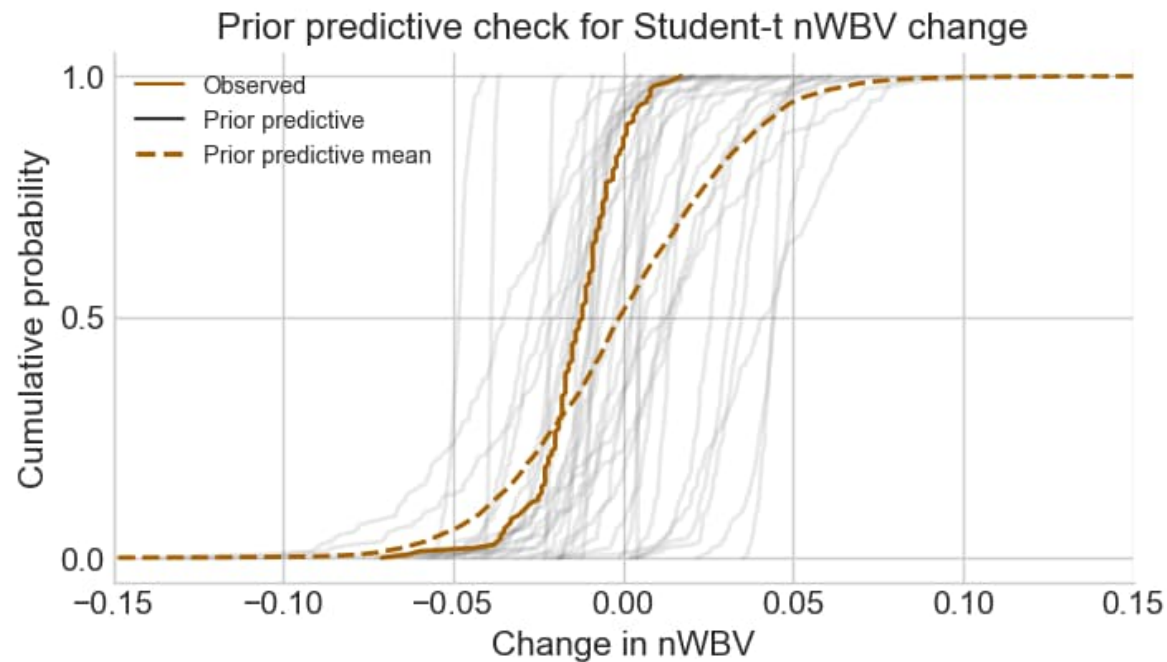
- The mean of the exponential is  $1/\lambda$ 
  - Bias towards smaller numbers
  - We chose it because we expected outliers
- Adding  $\nu_{\min}$  keeps the tails from becoming too wide
  - Too wide tails lead to unrealistic priors
  - And sampling issues

```
nu_min_student = 2  
lam_nu_student = 1 / 10
```

```
with pm.Model() as paired_student_t_model:  
    mu_change = pm.Normal("mu_change", mu=mu_mu_student, sigma=sigma_mu_student)  
    sigma_change = pm.HalfNormal("sigma_change", sigma=sigma_sigma_student)  
    nu_excess = pm.Exponential("nu_excess", lam=lam_nu_student)  
    nu = pm.Deterministic("nu", nu_min_student + nu_excess)  
    change_obs = pm.StudentT("change_obs", nu=nu, mu=mu_change, sigma=sigma_change, observed=change_paired)
```

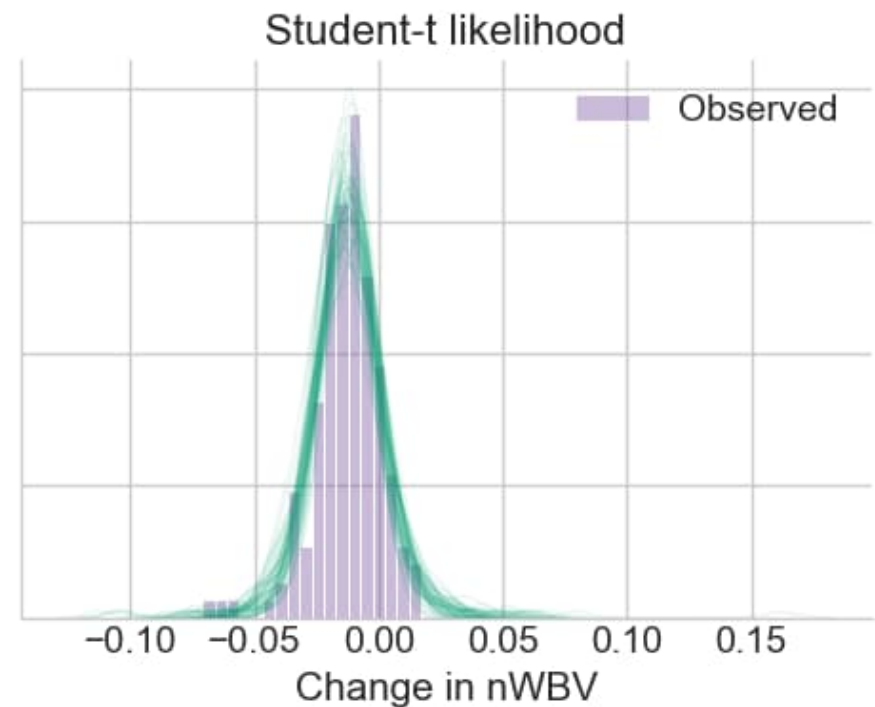
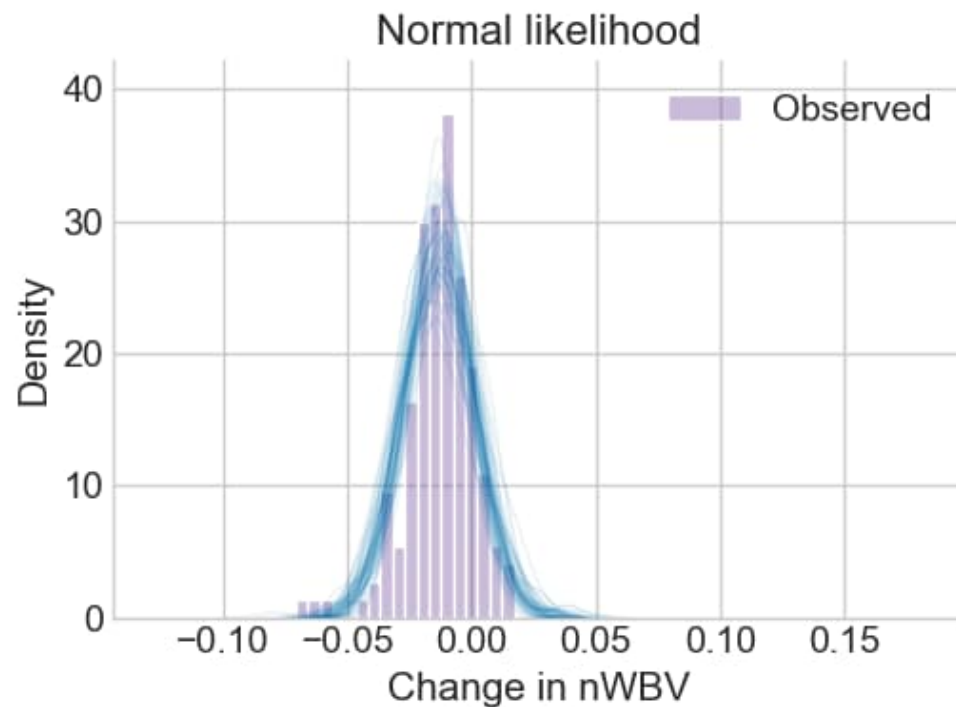
# Prior predictive and trace look reasonable

|              | mean   | sd    | hdi_3% | hdi_97% | mcse_mean | mcse_sd | ess_bulk | ess_tail | r_hat |
|--------------|--------|-------|--------|---------|-----------|---------|----------|----------|-------|
| mu_change    | -0.013 | 0.001 | -0.015 | -0.011  | 0.0       | 0.000   | 1489.0   | 1392.0   | 1.0   |
| sigma_change | 0.011  | 0.001 | 0.009  | 0.013   | 0.0       | 0.000   | 1416.0   | 1424.0   | 1.0   |
| nu           | 6.087  | 3.296 | 2.339  | 11.241  | 0.1       | 0.206   | 1206.0   | 1230.0   | 1.0   |



# Posterior predictive looks better

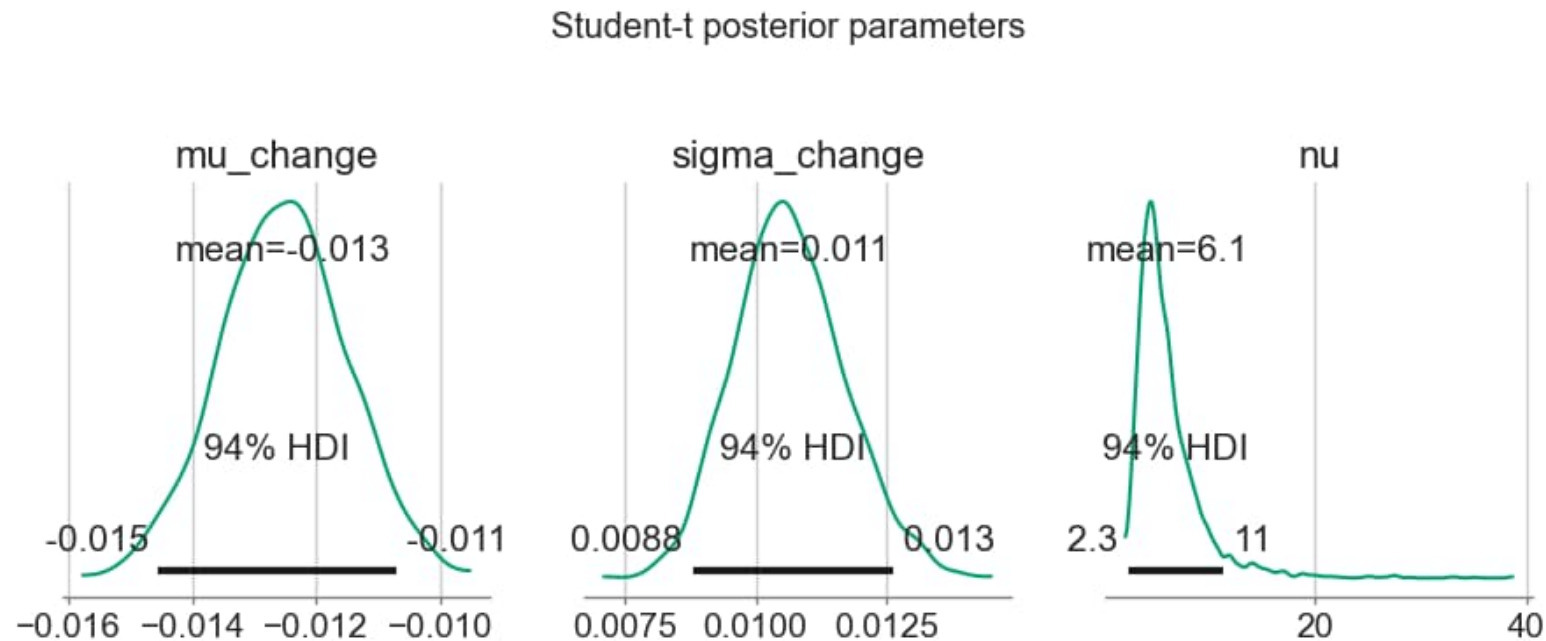
- The t distribution explains
  - wider tails
  - sharper peaks





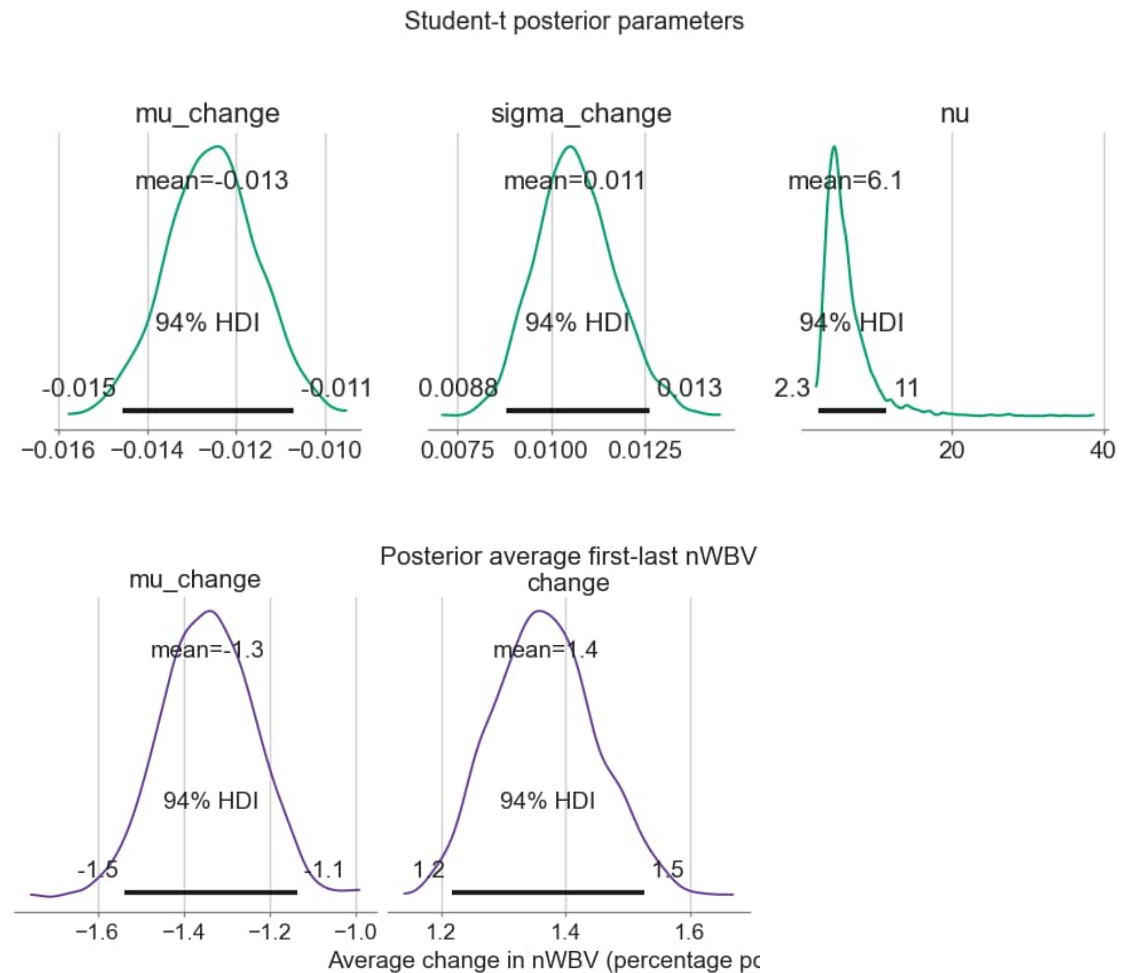
# Examine parameters

- The small values of  $\nu$  suggest that the normal was a bad fit



# Compare parameters to normal model fit

- $\mu$ 
  - Estimate similar
  - Less uncertainty
- $\sigma$ 
  - Estimate smaller
  - Less uncertainty

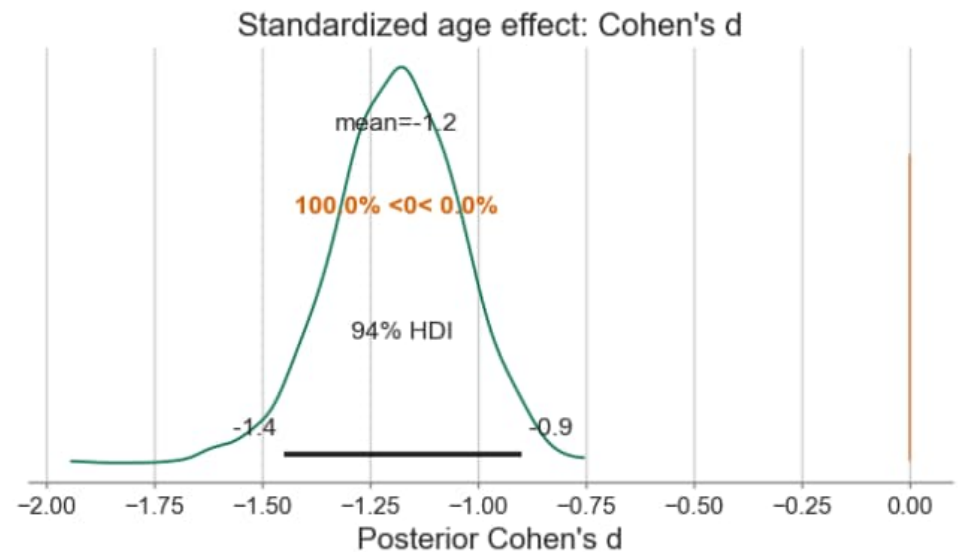
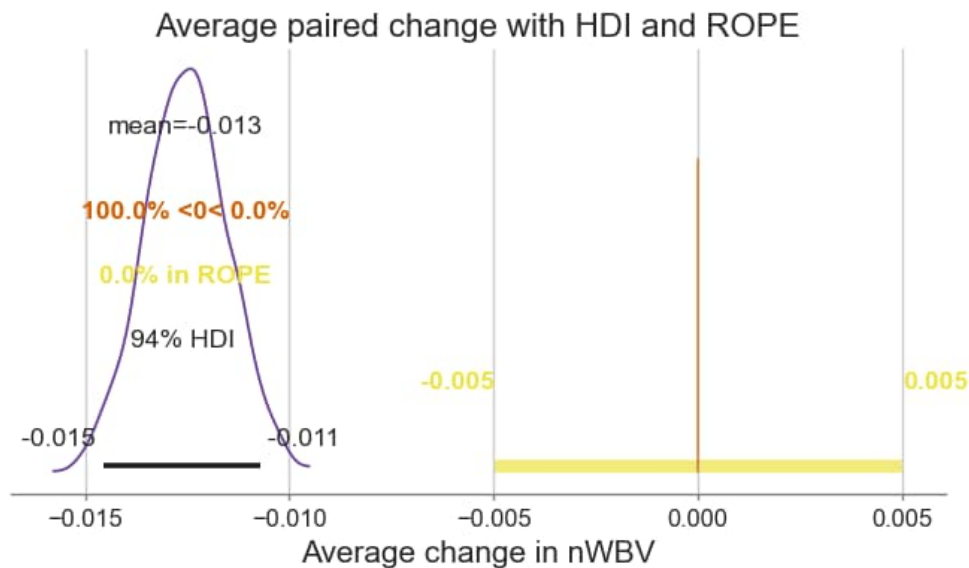


## 7.6 Effect sizes for paired comparisons

Similar to effect sizes for unpaired

# Effect size and HDI / ROPE test show large significant effect

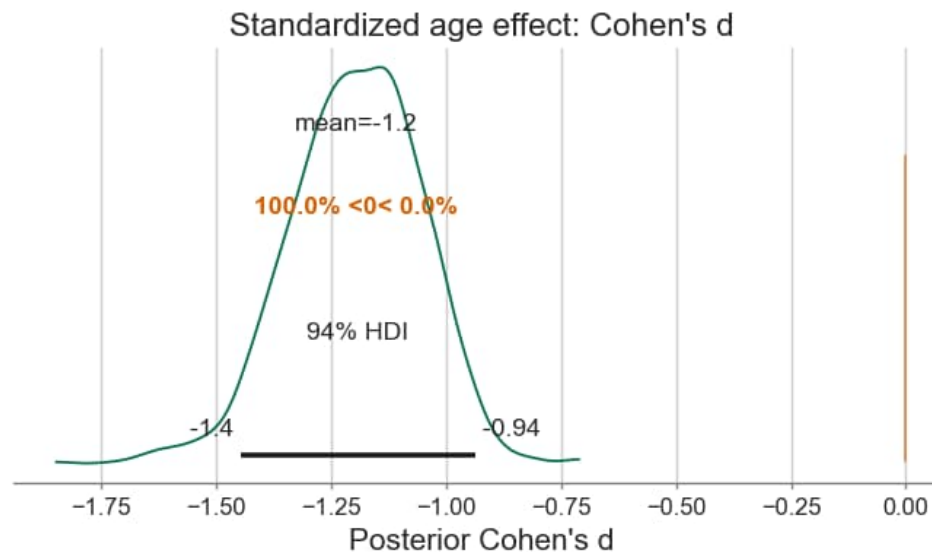
- 1% change in brain volume between first and last sessions
- Large effect size
- Far from ROPE



# Adding a calculation to the posterior after sampling

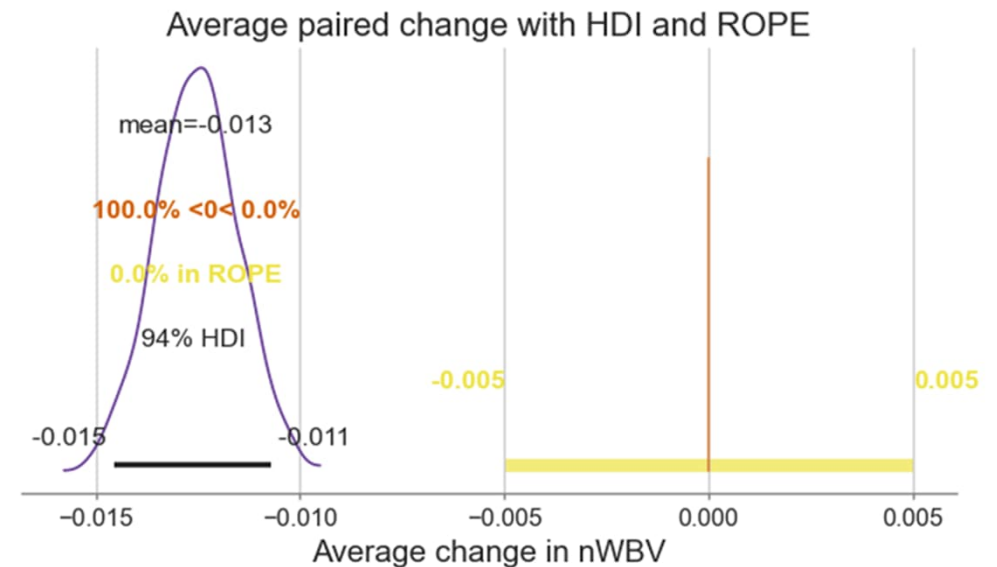
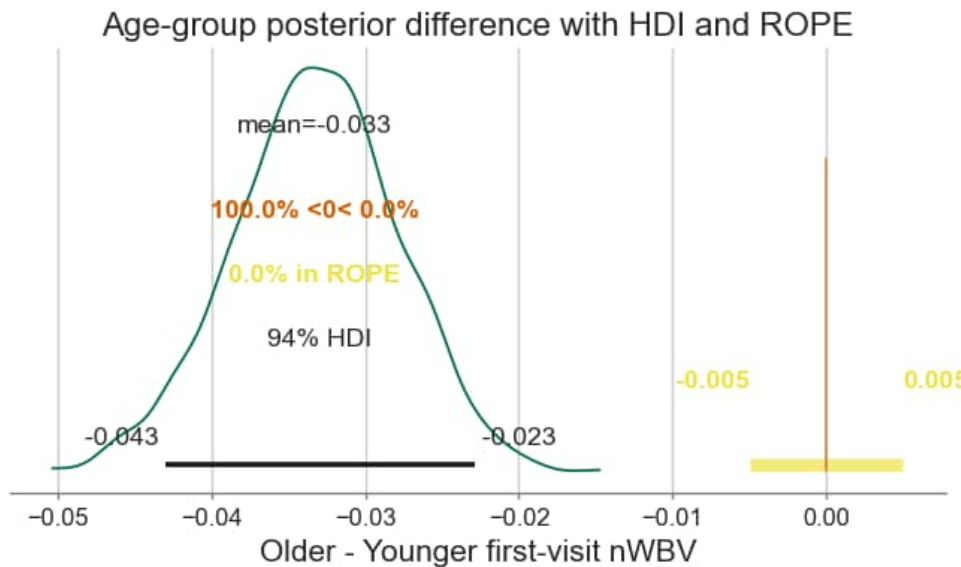
- These are operations on XArray DataArray objects
- Automatic handling of extra dimensions is called “broadcasting”
- Named dimensions help XArray broadcast correctly

```
idata.posterior["cohen_d"] = idata.posterior["mu_change"] / idata.posterior["sigma_change"]
```



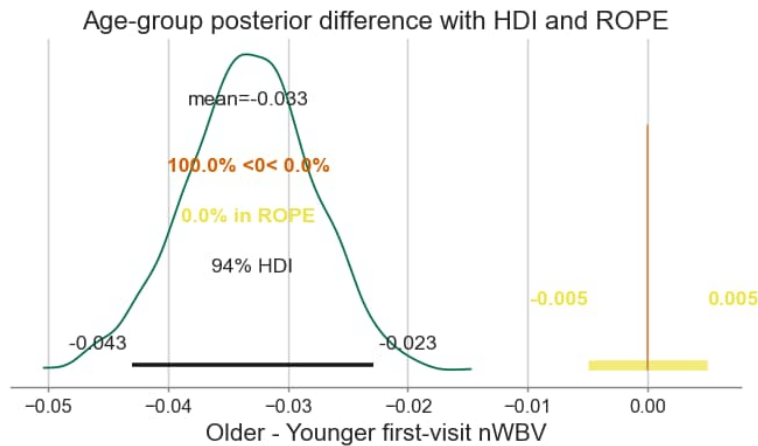
# Compare paired and unpaired analyses

- We can add a calculation to the posterior if we didn't sample it

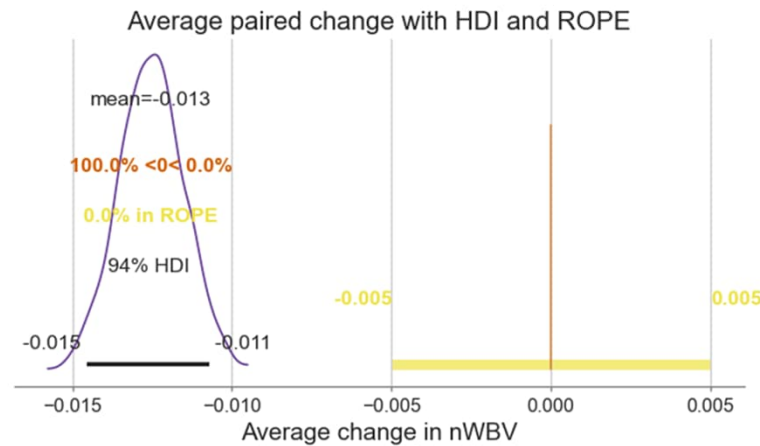


# Compare raw and normalized effects across analyses

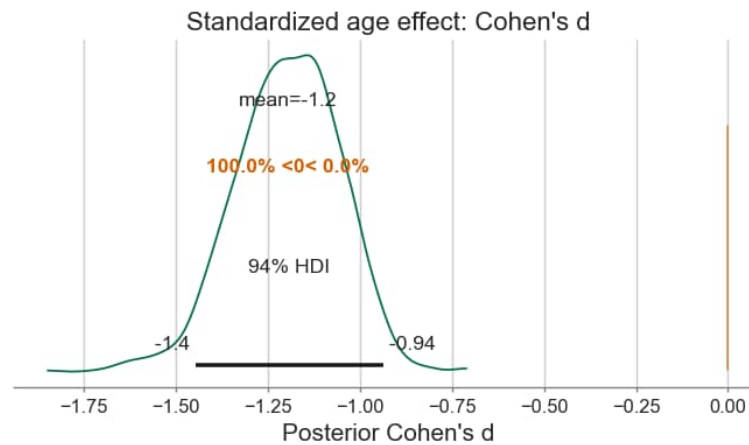
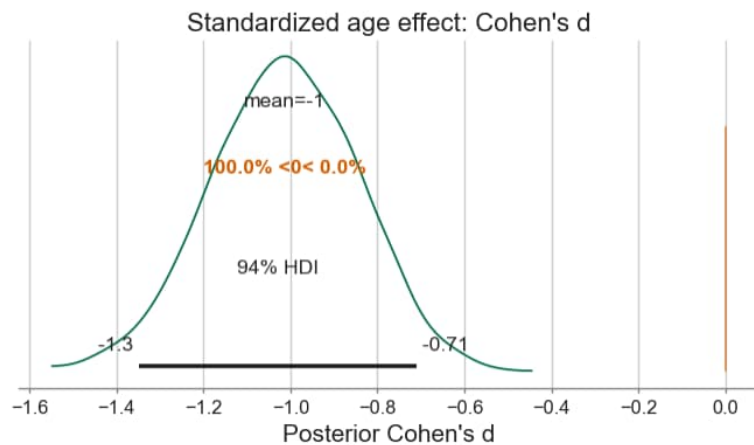
## Unpaired



## Paired



Raw effect



Normalized

# Tests on paired data

- The t-test assumes that the likelihood is normal
  - Not appropriate for non-normal data
- The Wilcoxon signed-rank test can be used

| Quantity                         | Value      |
|----------------------------------|------------|
| Sample mean of paired difference | -0.013400  |
| Sample SD of paired difference   | 0.013600   |
| Cohen's dz                       | -0.987500  |
| Degrees of freedom               | 149.000000 |
| 94% two-sided  t  threshold      | 1.895200   |
| Two-sided p-value                | 0.000000   |

| Quantity                        | Value       |
|---------------------------------|-------------|
| W statistic                     | 603.500000  |
| 94% two-sided W lower threshold | 4402.770200 |
| Two-sided p-value               | 0.000000    |



## 7.7 Next week: linear models

Linear models, or regression, are the workhorse of statistical analysis

# What we've done

- Data
  - And all of its complications
- Models
  - Combined with data, they tell us something about the world
- Parameters
  - Can be estimated once we have data
  - By finding their posterior distribution
- Complexity
  - Hierarchical modeling can express structure in the data

# Today and next week

- Up until now
  - Always only one variable
  - Except grouping variables
- From next week
  - How different variables are related to each other
    - Height and weight
    - Intelligence and brain size
    - Political affiliation and musical taste
    - Gene expression and risk of Parkinson's
    - Neural activity and intended direction of movement